Cointegration and error-correction models

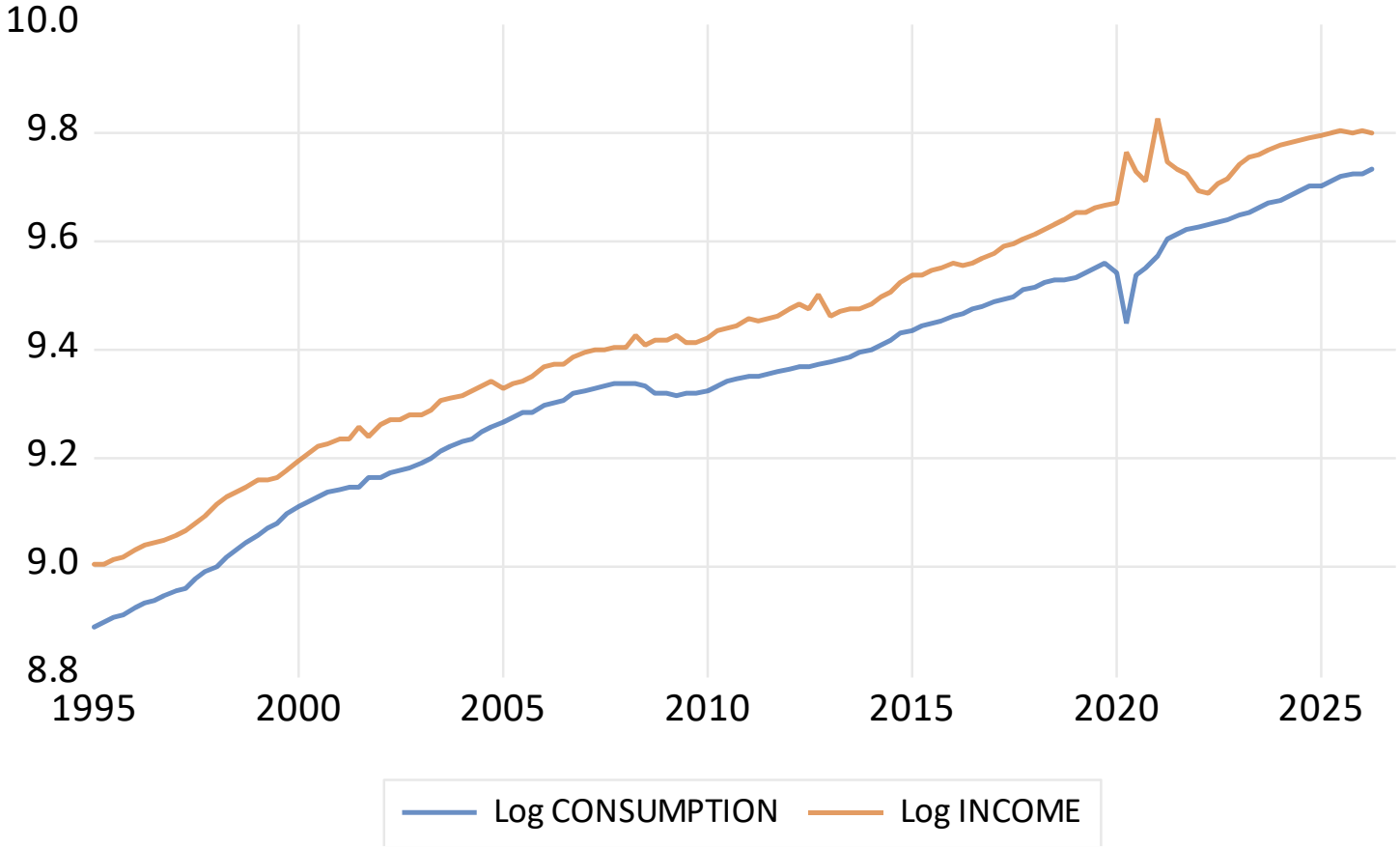
Econometrics. Lecture 14

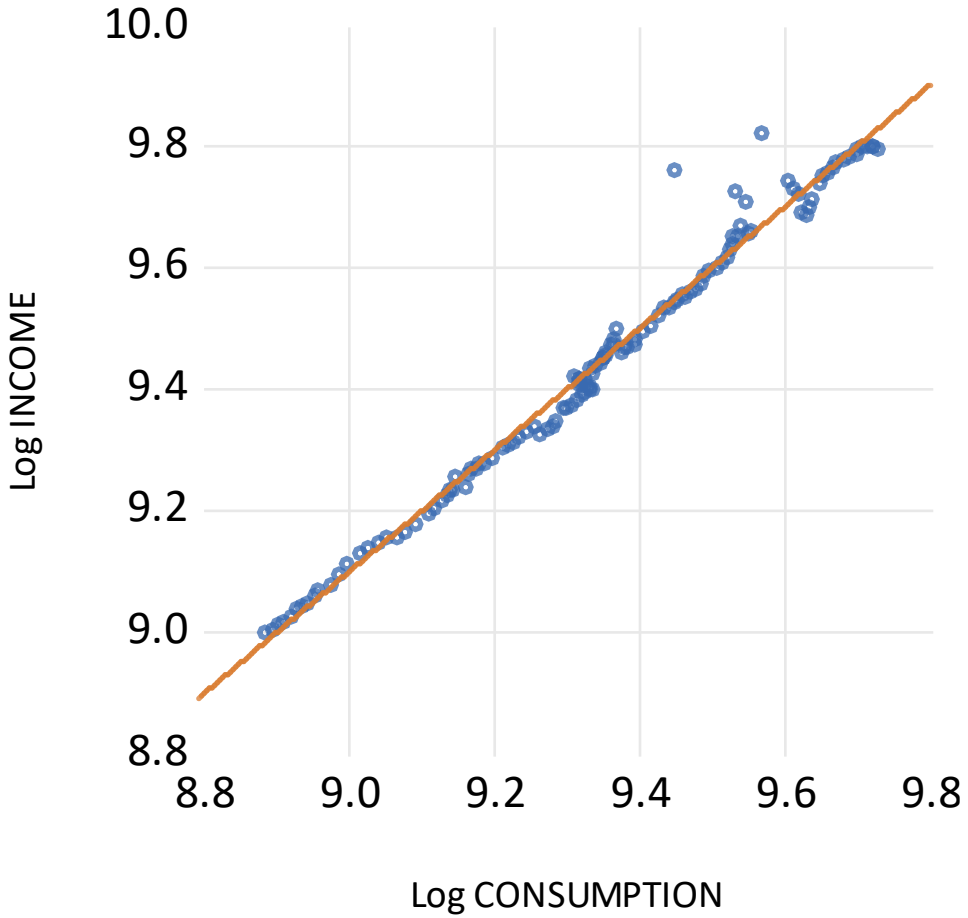
Content

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1. The concept of cointegration

Household income and consumption in US





Data from FRED:

- Real personal consumption expenditures, billions of chained 2017 dollars, seasonally adjusted.
- Real disposable personal income, billions of chained 2017 dollars, seasonally adjusted.

The problem

- As the time series are non-stationary and I(1), we can difference them to ensure stationarity of our variables:

$$\Delta y_t = b_0 + b_1 \Delta x_t + \Delta u_t.$$

- However, what we have from this equation is only the short-run relationship between the two variables.
- If there is a co-movement between y_t and x_t and they maintain an economically meaningful equilibrium in the long-run, we lose this long-term dynamics.
- But economists are interested in long-run relationships. Is there a supply and demand misalignment? Is the exchange rate over/undervalued? Is monetary policy loose or tight?
- The concept of cointegration and the error correction mechanism are very useful to resolve this problem.

When can a combination of I(1) series be stationary?

- Add two random walks together and you get a random walk. Differencing removes the trend but also removes the level information – the long run goes with it.
- The exception is when the series share the *same* stochastic trend. Then a particular weighted difference cancels it, leaving something stationary.
- Where would such a shared trend come from? Economics, not statistics.
- If a relationship holds in the long run, the gap between the two sides cannot wander freely – the theory *is* the restriction that makes the combination stationary.
- Standard examples: purchasing power parity, uncovered interest parity, the permanent income hypothesis, commodity market arbitrage.

What is cointegration?

Engle and Granger (1987) note that a linear combination of two or more I(1) series may be stationary, or I(0), in which case we say the series are **cointegrated**.

- For consumption and income in logs (c, r), cointegration means there exist b_1 and b_2 such that $b_1 c + b_2 r$ is stationary or I(0).
- If $b_1 c + b_2 r$ is stationary while *c* and *r* individually are not, the two series cannot drift apart without limit. Something has to pull them back.
- That "something" is the subject of the rest of the lecture: an adjustment mechanism keeping deviations from $b_1 c + b_2 r = 0$ bounded rather than explosive.

The error-correction mechanism

- In the long run, the consumption depends on the income:

$$c_t = \beta_0 + \beta_1 r_t + u_t$$ where c_t is log of consumption and r_t is the log of income.

- At any given time *t*, you can consume more (for example, by taking out a loan) or less (for example, due to expectations of a crisis) than expected based on your income. And this **disequilibrium** is captured by the **error term** u_t.

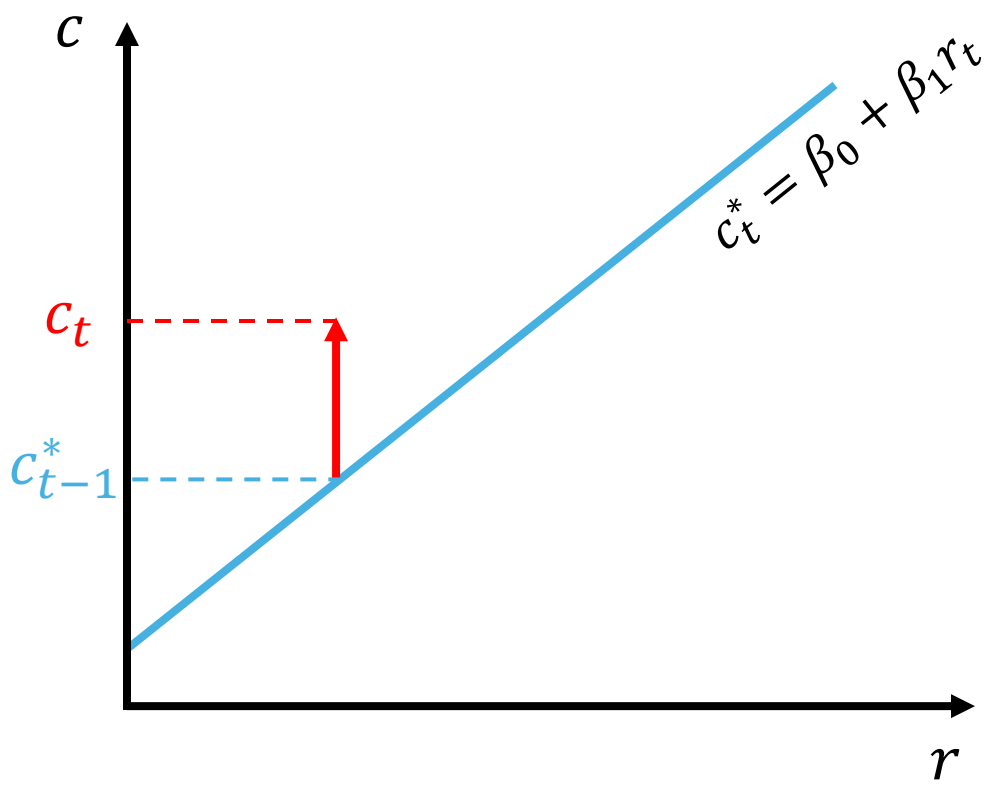
- The deviation of consumption from income cannot be infinite: you will have to pay off the loan in the future (by reducing purchases of other goods and services). Therefore, in the future, consumption will be adjusted to the level that corresponds to the level of income. This is how the **error-correction** (or equilibrium) **mechanism** works.

Off the long-run line: what now?

- Suppose that in the long run c and r are cointegrated:

$c_t = \beta_0 + \beta_1 r_t + u_t$ where $\beta_1 > 0$.

- Suppose $c \uparrow$ such that $c_t - \beta_1 r_t > 0$.
- What would be the dynamics?

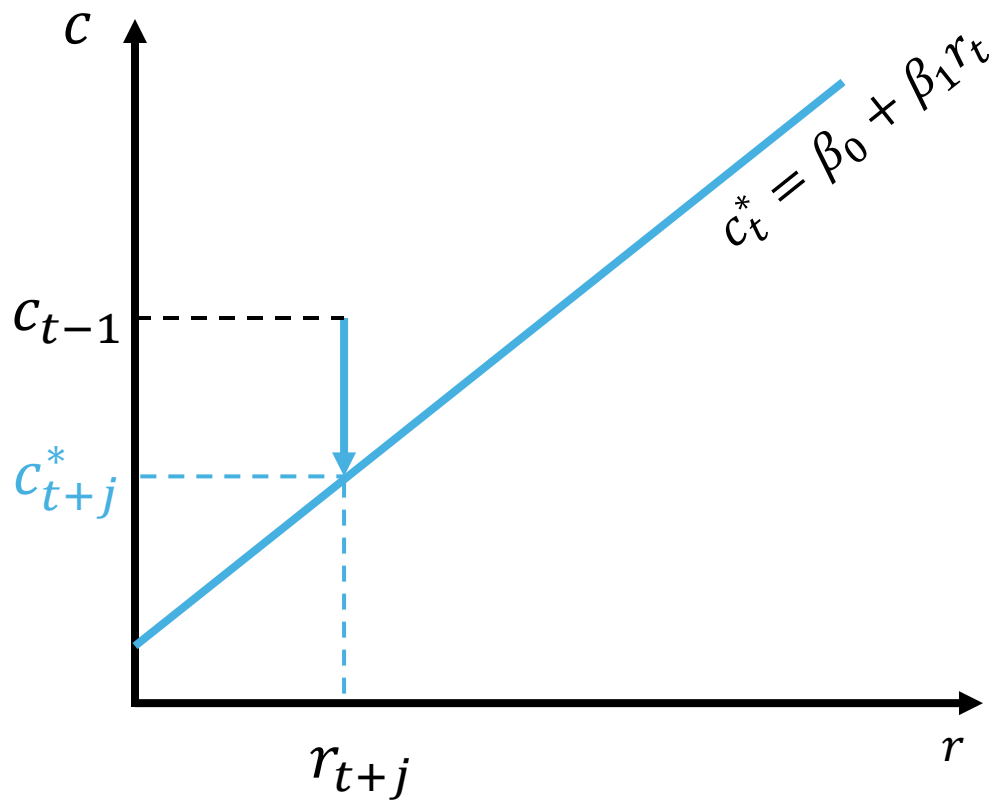


Case 1: c_t is doing all the adjustment

- c_t^* is unchanged and $c_t \downarrow$:

$\Delta c_t = \alpha_c(c_{t-1} - c_{t-1}^*)$ where $\alpha_c < 0$.

- The change in c_t this period is proportional to last period's gap – the further from the line, the harder the pull.
- α_c is the error-correction coefficient. It is the **speed of return to equilibrium** – its negative sign guarantees the pull is corrective, and its magnitude says **how much of the gap closes each period.**

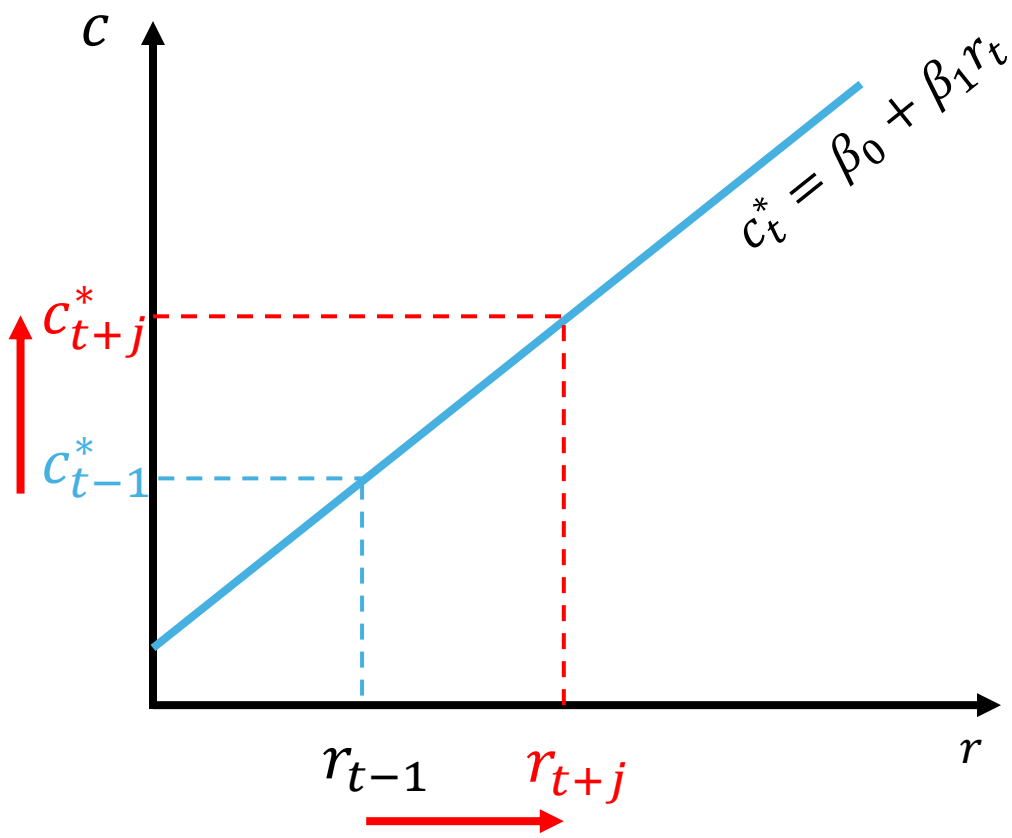


Case 2: r_t (and hence c_t^*) is doing all the adjustment

- c_t is unchanged, r_t and c_t^* ↑:

$\Delta r_t = \alpha_r(c_{t-1} - c_{t-1}^*)$ where $\alpha_r > 0$.

- Short run change in r_t is a linear function of the deviation from the long run equilibrium.



Case 3: both c_t and r_t are adjusting

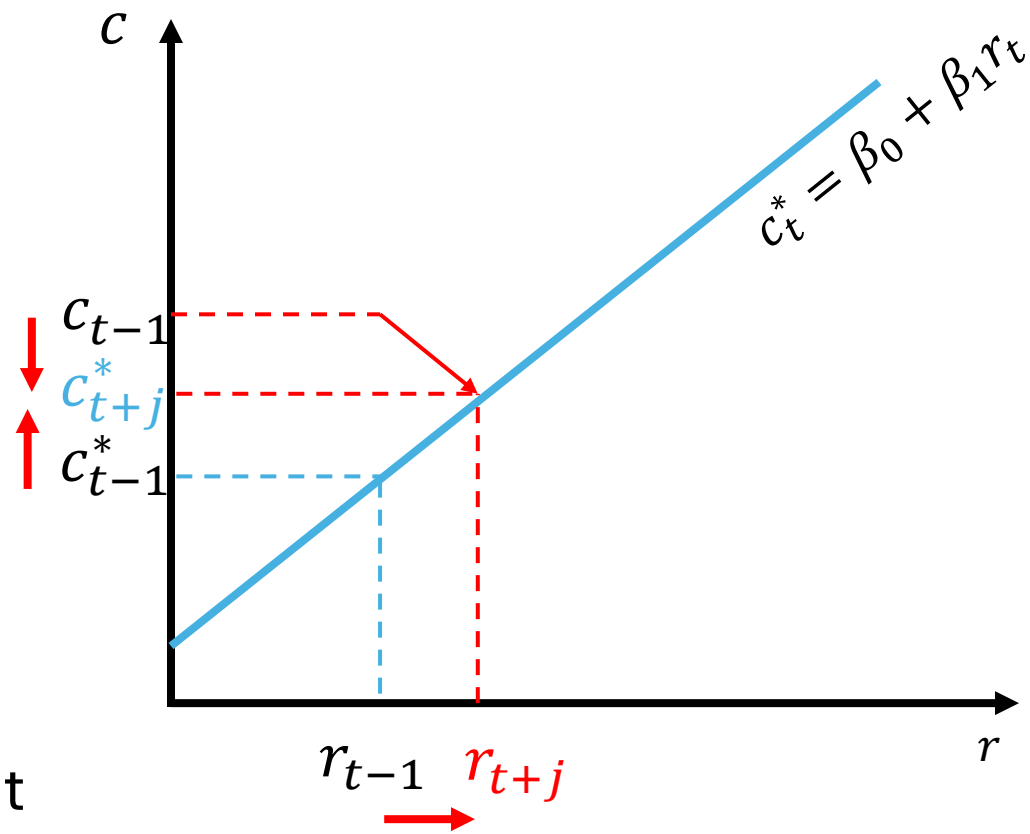
- c_t and r_t adjusting simultaneously:

$$\Delta c_t = \alpha_c(c_{t-1} - c^*_{t-1}) \quad \text{where } \alpha_c < 0,$$

$$\Delta r_t = \alpha_r(c_{t-1} - c^*_{t-1}) \quad \text{where } \alpha_r > 0.$$

- The magnitude of coefficients α_c and α_r determines how fast the system returns to the long-run relationship.

- This is a basic error-correction model. It generalises to three or more variables.



The error-correction model (ECM)

- If y_t and x_t are cointegrated, their linear combination $\hat{u}_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t$ is stationary. So, the residuals by definition $\hat{u}_t \sim I(0)$.
- Thus, we can express the relationship between y_t and x_t with an error-correction model specification as:

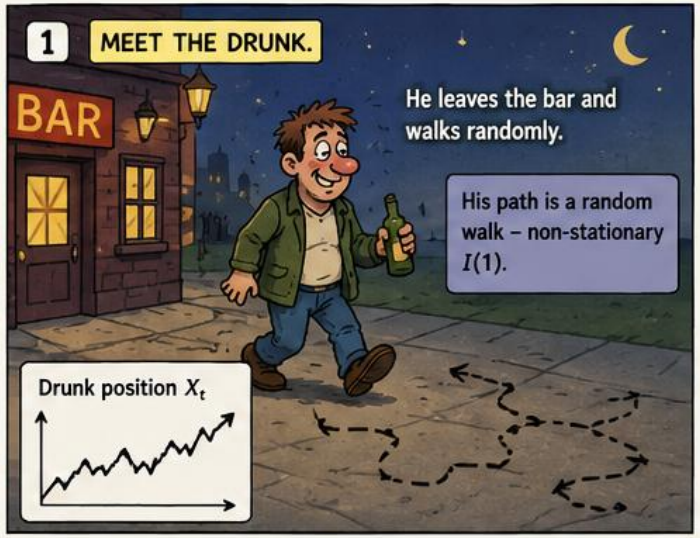
$$\Delta y_t = b_0 + b_1 \Delta x_t + \alpha(\hat{u}_{t-1}) + e_t, \qquad \text{where } \alpha < 0$$

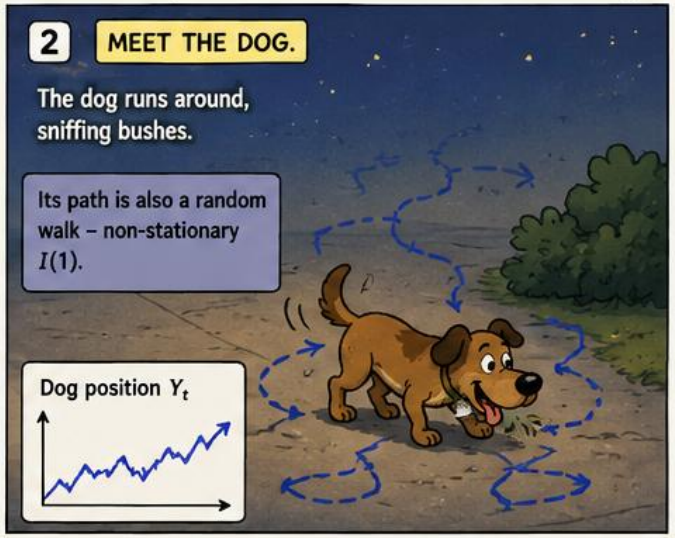
  which will now have the advantage of including both long-run and short-run information

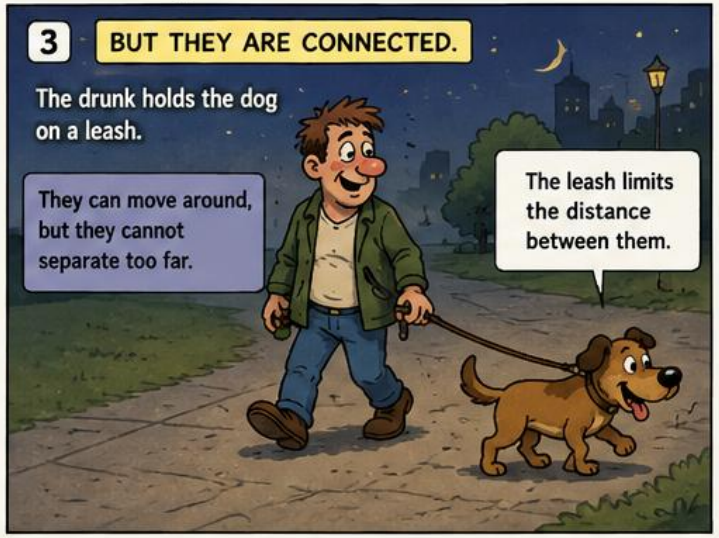
- b_1 is the impact multiplier or the short-run effect of Δx_t on Δy_t.
- α is the error-correction coefficient, and shows how much of the disequilibrium is being corrected in period t.
- We can include lags of Δy_t and Δx_t in the ECM or other exogenous variables.

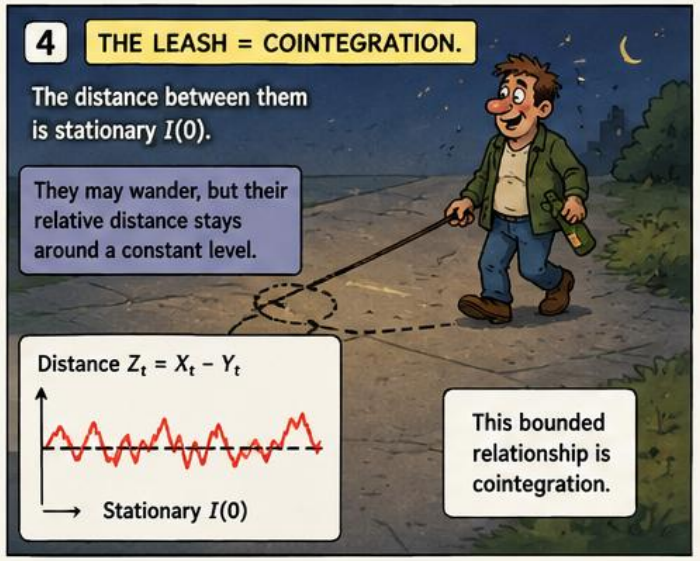
COINTEGRATION: THE DRUNK AND HIS DOG

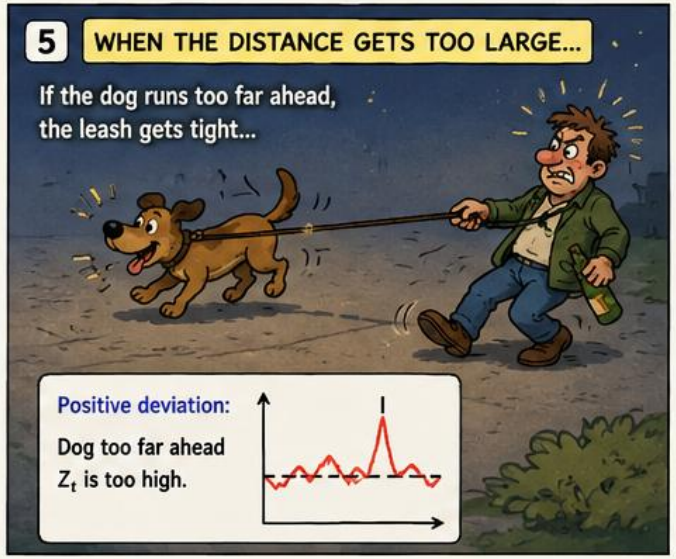
Two non-stationary series, but a stationary relationship.

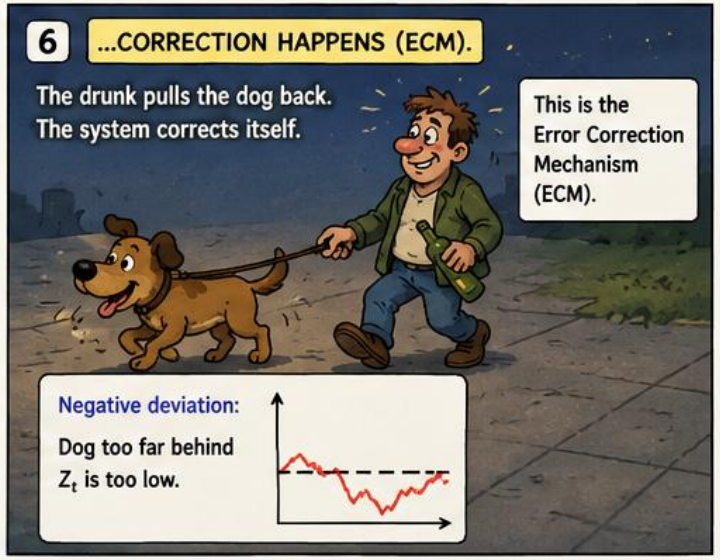












THE BIG IDEA:
Two non-stationary series X_t and Y_t can be cointegrated if a linear combination (e.g., $X_t - Y_t$) is stationary.
They move randomly, but stay tied together by an economic relationship (the leash).

$$X_t \sim I(1),\quad Y_t \sim I(1),\quad X_t - Y_t \sim I(0)$$

That's cointegration.

A Drunk and Her Dog

Murray, M. P. (1994). A Drunk and Her Dog: An Illustration of Cointegration and Error Correction. *The American Statistician*, *48*(1), 37–39. https://doi.org/10.2307/2685084

Summary

- Two unrelated trending variables can have high correlation. But their linear combination is I(1) – **spurious regression**, which is why the regression between them produces significant t-statistics that mean nothing – the spurious regression problem.
- **Cointegration** is stronger:
  - not just moving together, …
  - … but **maintaining an economically meaningful equilibrium** relation over time.
- Cointegration and **error correction** are two descriptions of the same thing: if a stationary combination exists, some adjustment mechanism must be enforcing it.

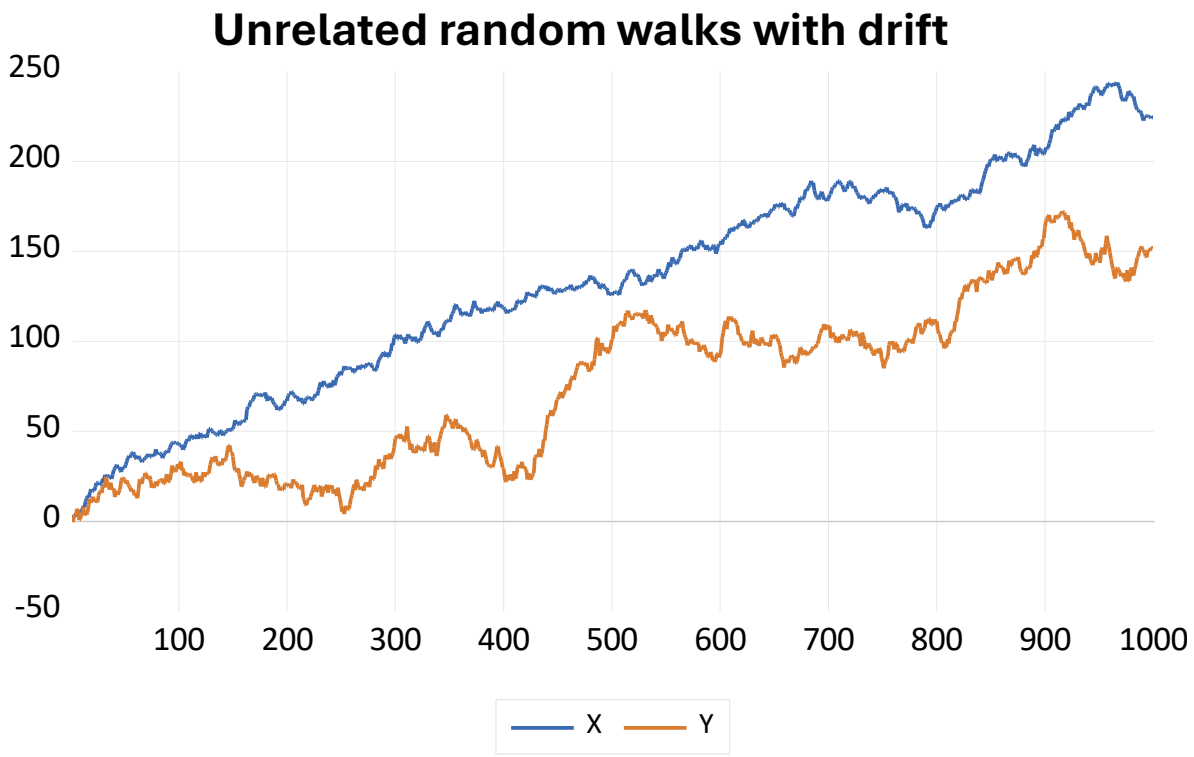
2. Error-correction mechanism: simulated data

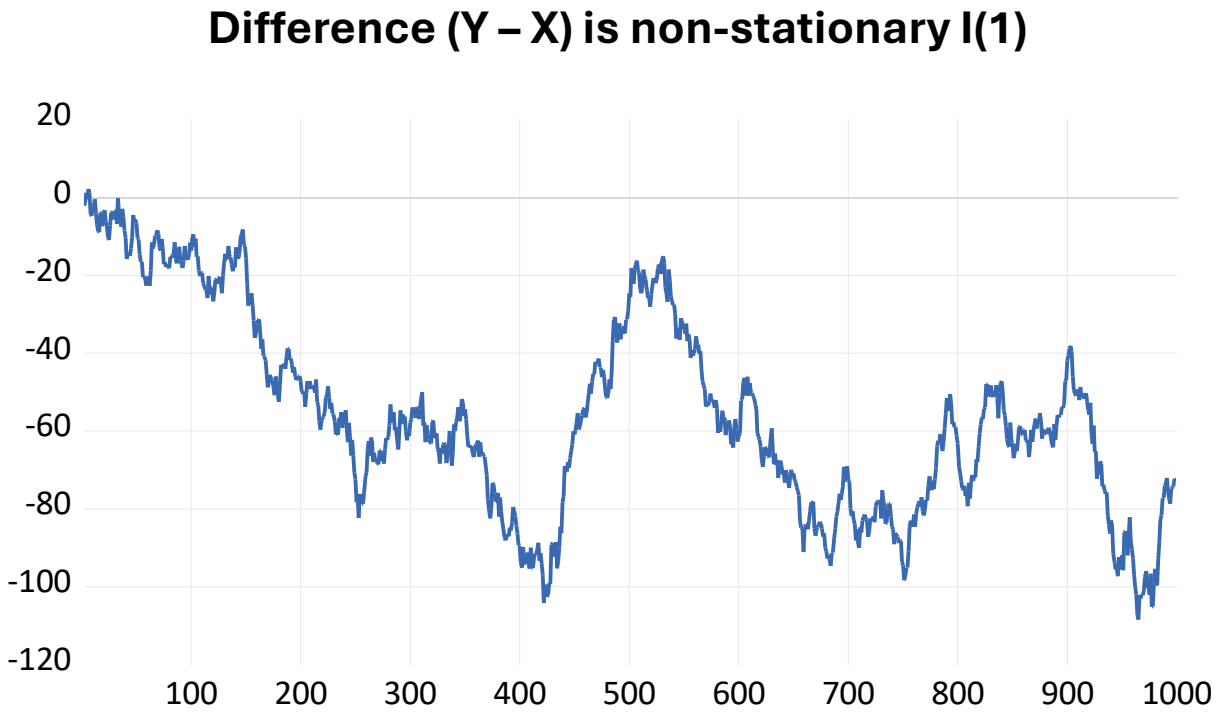
Unrelated random walks: no long-run equilibrium

Generated data:

x = x(-1) + 0.20 + 1.5*nrnd

y = y(-1) + 0.05 + 2.5*nrnd





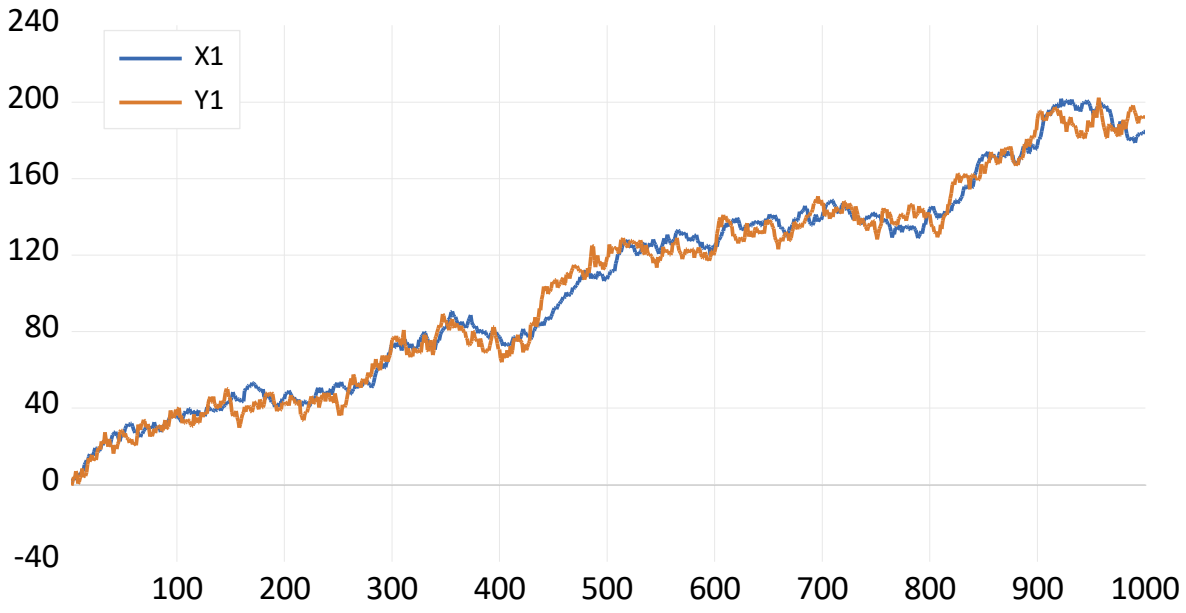
Cointegrated random walks: well-defined long-run equilibrium

Generated data:

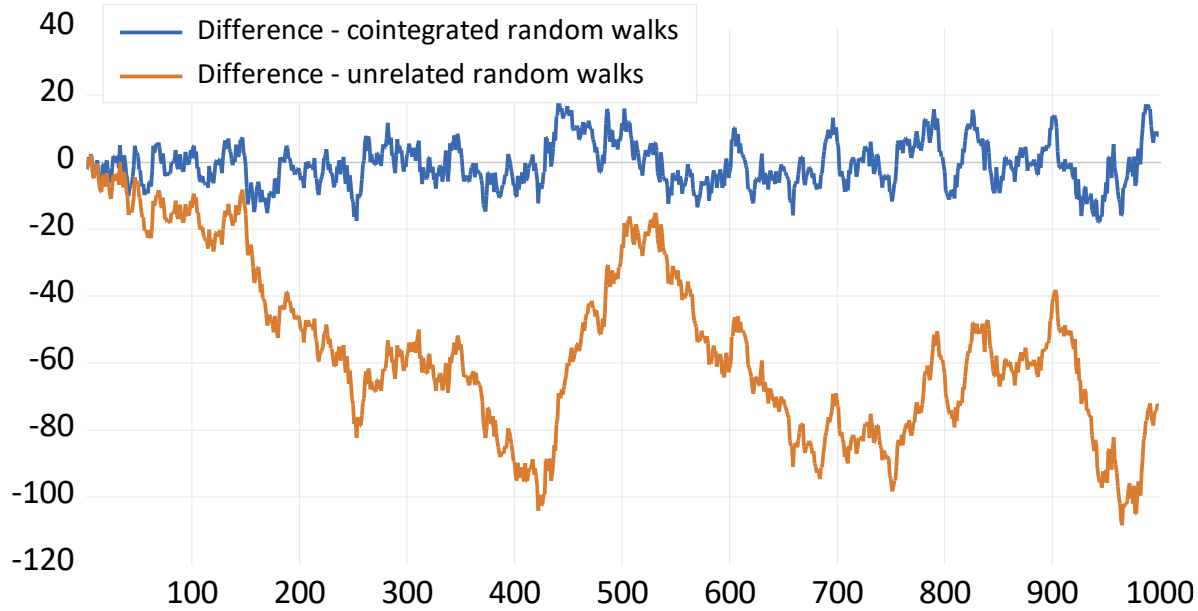
x1 = x1(-1) + 0.20 + 1.5*nrnd **+ 0.05*(y1(-1) – x1(-1))**

y1 = y1(-1) + 0.05 + 2.5*nrnd **– 0.05*(y1(-1) – x1(-1))**

Cointegrated random walks with drift



Difference (Y1 – X1) is stationary I(0)



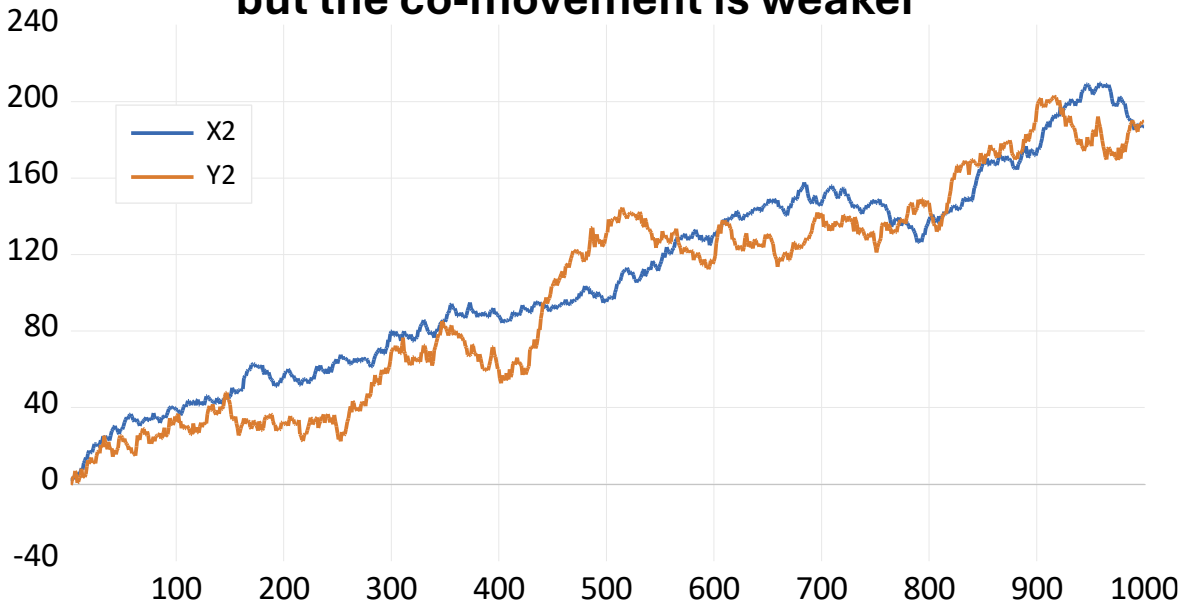
Magnitude of the error-correction coefficient: smaller value weakens the co-movement

Generated data:

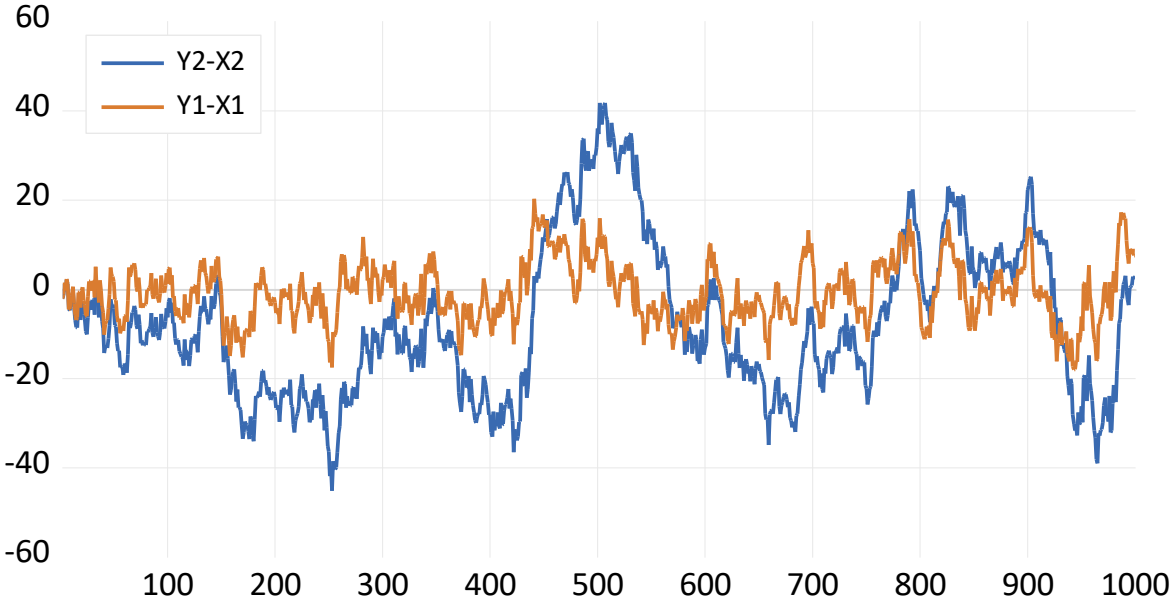
x2 = x2(-1) + 0.20 + 1.5*nrnd **+ 0.005*(y2(-1) – x2(-1))**

y2 = y2(-1) + 0.05 + 2.5*nrnd **– 0.005*(y2(-1) – x2(-1))**

Random walks are still cointegrated, but the co-movement is weaker



Difference (Y2 – X2) is still stationary I(0), but the adjustment process is slower



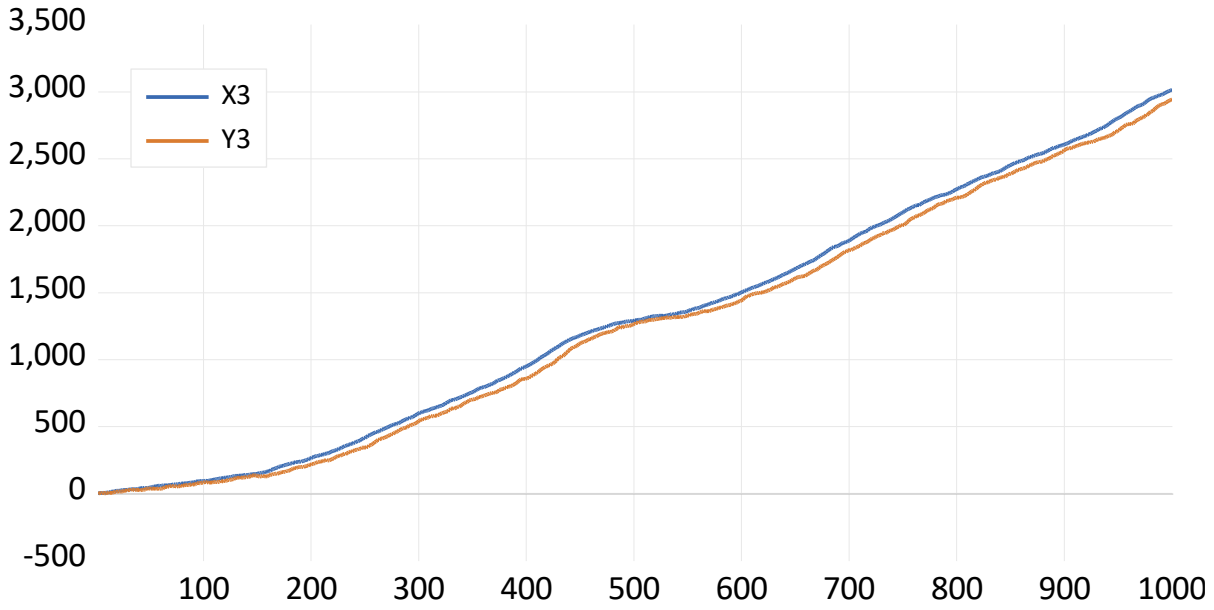
Signs of the error-correction coefficients: they must pull in opposite directions

Generated data:

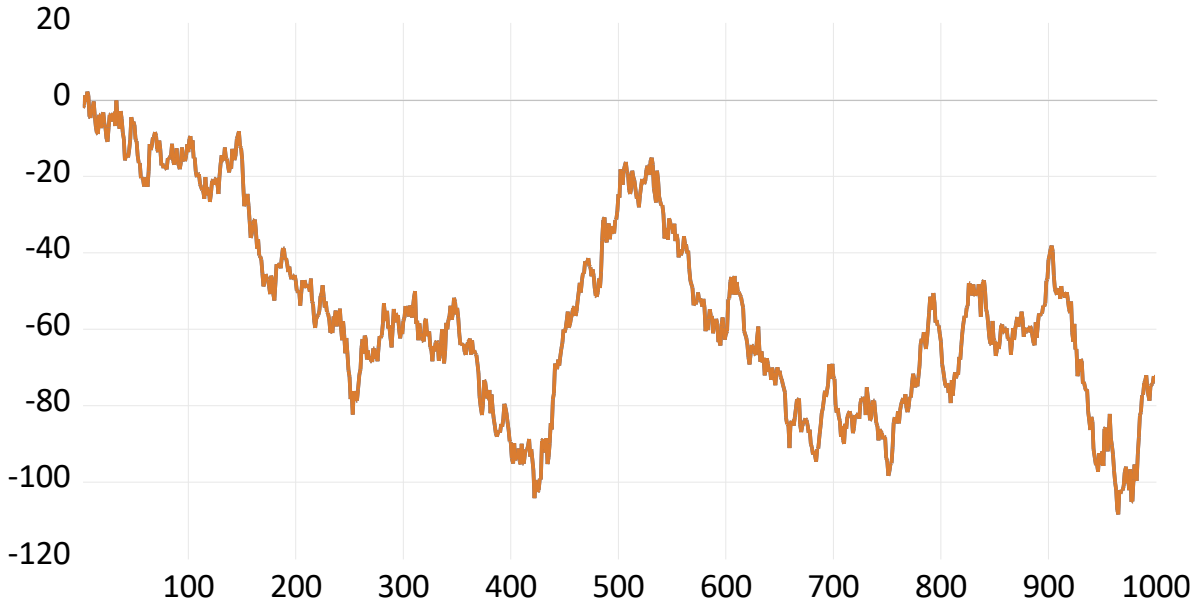
x3 = x3(-1) + 0.20 + 1.5*nrnd **– 0.05*(y3(-1) – x3(-1))**

y3 = y3(-1) + 0.05 + 2.5*nrnd **– 0.05*(y3(-1) – x3(-1))**

Random walks are not cointegrated



Difference (Y3 – X3) is non-stationary I(1)



Summary

- Two independent random walks diverge.
- An error-correction model with the correct signs produces a stationary distance and hence cointegration.
- An error-correction model with the wrong signs produces a non-stationary distance and no cointegration.
- An error-correction coefficient is the speed of return to equilibrium – it shows how much of the gap closes each period.

3. Cointegration in single equations: the Engle–Granger approach

Step 1.
Test the variables for their order of integration

- Cointegration requires that the variables be integrated of the same order. Apply the unit-root test (ADF, PP) or the stationarity (KPSS) tests to determine the order of integration:
  - both variables are stationary I(0) – estimate the classical regression or VAR in levels. I(0) series already reverts to its mean on its own;
  - the variables are integrated of different order – no cointegration;
  - both variables are integrated of the same order (almost always I(1)) – cointegration and the error-correction mechanism may be present; we proceed with step 2.

Step 2.
Estimate the long-run relationship

- If the results of step 1 indicate that both y_t and x_t are integrated of the same order (usually in economics, I(1)), the next step is to estimate the long-run equilibrium relationship of the form

$$y_t = \beta_0 + \beta_1 x_t + u_t,$$

  and obtain the residuals $\hat{u}_t$ of this equation.

- If there is no cointegration, the results obtained will be spurious.
- If the variables are cointegrated, then OLS regression yields 'super-consistent' estimators for the cointegrating parameter $\hat{\beta}_1$.

Step 3.
Test if the residuals are stationary

- To determine if the variables are in fact cointegrated, the estimated residuals $\hat{u}_t$ of the long-run relationship (or the deviations from long-run equilibrium) must be stationary.
- Perform the ADF test on the residual series $\hat{u}_t$. Do not include a constant or a time trend.
- The critical values for the null of no cointegration differ from the standard ADF values, being more negative (typically around –3.5; a more comprehensive set of critical values may be found in MacKinnon (2010)).
- If we find that $\hat{u}_t \sim I(0)$, we can reject the null that the variables y_t and x_t are not cointegrated (similarly, if we have a single equation with more than just one explanatory variable).

Step 4.
Estimate the error-correction model (ECM)

- Estimate the linear regression model for the changes of y_t and x_t adding the lagged residuals as an explanatory variable (add lags of Δy_t and/or Δx_t if needed):

$$\Delta y_t = b_0 + b_1 \Delta x_t + \alpha(\hat{u}_{t-1}) + e_t, \qquad \text{where } \alpha < 0$$

- Coefficient on the lagged residual terms should be statistically significant and has a correct sign to ensure that the error-correction mechanism works.
- The adequacy of the model must always be checked by performing diagnostic tests: residuals of the ECM should be white-noise (not correlated, homoscedastic and, ideally, normally distributed).

Drawbacks of the Engle-Granger approach

- The EG approach does not say anything about which of the variables can be used as a regressor and why. One can either regress y_t on x_t or choose to reverse the order and regress x_t on y_t. The problem becomes more complicated when there are more than two variables to test. Use the economic theory to decide.
- When there are more than two variables there may be more than one cointegrating relationship, and the EG procedure cannot treat this possibility. Use Johansen's (1991; 1995) approach if you have reason to believe that there are multiple cointegrating relationships (not covered but can be estimated in EViews).
- EG relies on a two-step estimator. The first step is to generate the residuals and the second is to estimate a regression for this series to see whether the series is stationary or not. Any error introduced in the first step is carried into the second.

The Engle-Granger approach considerations. FMOLS/DOLS instead of OLS in Step 2

- The problem: in a cointegrating (long-run) regression, **OLS** gives a **consistent** β but **invalid standard errors**:
  - OLS estimate of β is fine (super-consistent – it even converges faster than usual);
  - OLS *inference* is not: the long-run error term is generally correlated with its own past (serial correlation) and often with the regressors (endogeneity), so *t*-stats and standard errors are biased.
- There are methods for estimating a single cointegrating vector, that correct this bias: Fully Modified OLS (**FMOLS**), Dynamic OLS (**DOLS**), and Canonical Cointegrating Regression (**CCR**).
  - FMOLS/DOLS/CCR fix the standard errors so you can actually test hypotheses about the long-run relationship.
- Use FMOLS or DOLS when you want to ***test*** the long-run coefficient, ***not just estimate*** it.

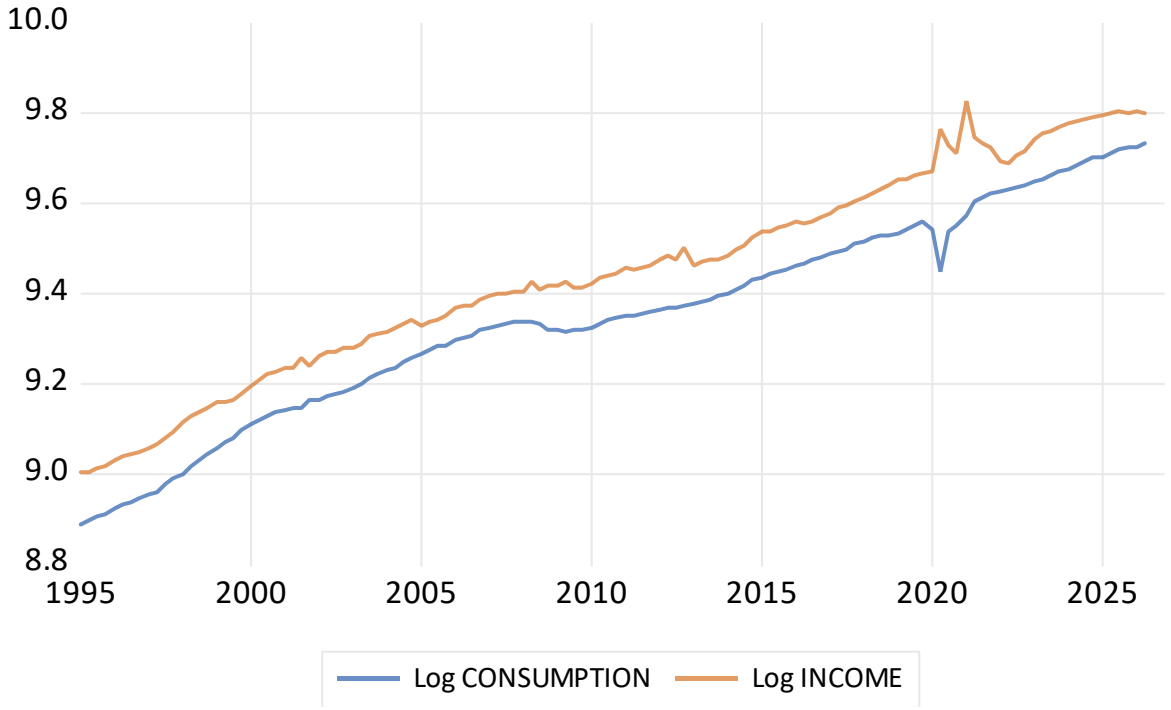
The Engle-Granger approach considerations. VAR(*p*) instead of single-equation regression in Step 4

- A single-equation ECM assumes **only one variable adjusts to disequilibrium** – it forces you to *assume* who adjusts.
- But we cannot be completely sure that it is this variable that is being adjusted: other variables may also adjust in response to deviations from equilibrium, or all variables may adjust.
- A system (**VAR**) lets **every variable adjust** – and lets you test which ones actually do.
- If several variables adjust, the single equation is misspecified; only the system captures the full adjustment.

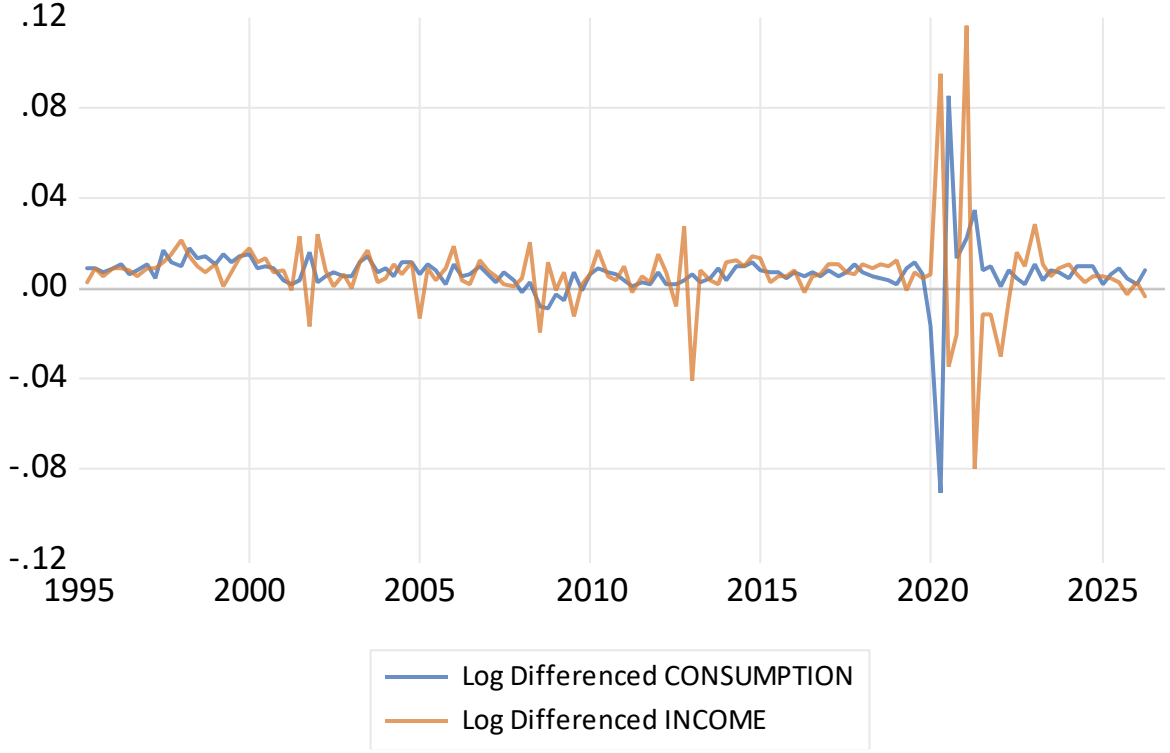
4. The Engle–Granger approach in EViews

Data.
Household income and consumption in US

Time series are non-stationary – have a stochastic trend



First differences (proxies for growth rates) look stationary



Data from FRED:

- Real personal consumption expenditures, billions of chained 2017 dollars, seasonally adjusted.
- Real disposable personal income, billions of chained 2017 dollars, seasonally adjusted.

Step 1. ADF test: time series are non-stationary in log-levels

Log consumption is non-stationary

Augmented Dickey-Fuller Unit Root Test on L_CONS

Null Hypothesis: L_CONS has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.572931	0.2934
Test critical values:	1% level	-4.033108	
	5% level	-3.446168	
	10% level	-3.148049	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(L_CONS)
Method: Least Squares
Date: 08/12/26 Time: 16:48
Sample (adjusted): 1995Q2 2026Q2
Included observations: 125 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
L_CONS(-1)	-0.077450	0.030102	-2.572931	0.0113
C	0.702398	0.269703	2.604336	0.0103
@TREND("1995Q1")	0.000441	0.000185	2.388023	0.0185

R-squared	0.057479	Mean dependent var	0.006753
Adjusted R-squared	0.042027	S.D. dependent var	0.012577
S.E. of regression	0.012310	Akaike info criterion	-5.933079
Sum squared resid	0.018488	Schwarz criterion	-5.865200
Log likelihood	373.8175	Hannan-Quinn criter.	-5.905504
F-statistic	3.720020	Durbin-Watson stat	2.275372
Prob(F-statistic)	0.027025		

Log income is non-stationary

Augmented Dickey-Fuller Unit Root Test on L_INC

Null Hypothesis: L_INC has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 3 (Automatic - based on SIC, maxlag=12)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.160360	0.0975
Test critical values:	1% level	-4.034997	
	5% level	-3.447072	
	10% level	-3.148578	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(L_INC)
Method: Least Squares
Date: 08/12/26 Time: 16:48
Sample (adjusted): 1996Q1 2026Q2
Included observations: 122 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
L_INC(-1)	-0.158929	0.050288	-3.160360	0.0020
D(L_INC(-1))	-0.269603	0.090344	-2.984181	0.0035
D(L_INC(-2))	-0.028561	0.092836	-0.307654	0.7589
D(L_INC(-3))	0.345507	0.085595	4.036537	0.0001
C	1.447498	0.454839	3.182438	0.0019
@TREND("1995Q1")	0.000936	0.000312	2.998079	0.0033

R-squared	0.315254	Mean dependent var	0.006400
Adjusted R-squared	0.285739	S.D. dependent var	0.018266
S.E. of regression	0.015437	Akaike info criterion	-5.456163
Sum squared resid	0.027643	Schwarz criterion	-5.318260
Log likelihood	338.8259	Hannan-Quinn criter.	-5.400151
F-statistic	10.68119	Durbin-Watson stat	1.997499
Prob(F-statistic)	0.000000		

Step 1. ADF test: time series are stationary in log-differences – I(1)

First difference of log consumption is stationary

Augmented Dickey-Fuller Unit Root Test on D(L_CONS)

Null Hypothesis: D(L_CONS) has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-12.96999	0.0000
Test critical values:	1% level	-3.483751	
	5% level	-2.884856	
	10% level	-2.579282	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(L_CONS,2)
Method: Least Squares
Date: 08/12/26 Time: 16:55
Sample (adjusted): 1995Q3 2026Q2
Included observations: 124 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(L_CONS(-1))	-1.159187	0.089375	-12.96999	0.0000
C	0.007810	0.001275	6.123354	0.0000

R-squared	0.579630	Mean dependent var	-7.71E-06
Adjusted R-squared	0.576185	S.D. dependent var	0.019227
S.E. of regression	0.012517	Akaike info criterion	-5.907464
Sum squared resid	0.019114	Schwarz criterion	-5.861976
Log likelihood	368.2628	Hannan-Quinn criter.	-5.888985
F-statistic	168.2206	Durbin-Watson stat	2.008646
Prob(F-statistic)	0.000000		

First difference of log income is stationary

Augmented Dickey-Fuller Unit Root Test on D(L_INC)

Null Hypothesis: D(L_INC) has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=12)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-5.543570	0.0000
Test critical values:	1% level	-3.484653	
	5% level	-2.885249	
	10% level	-2.579491	

*MacKinnon (1996) one-sided p-values.

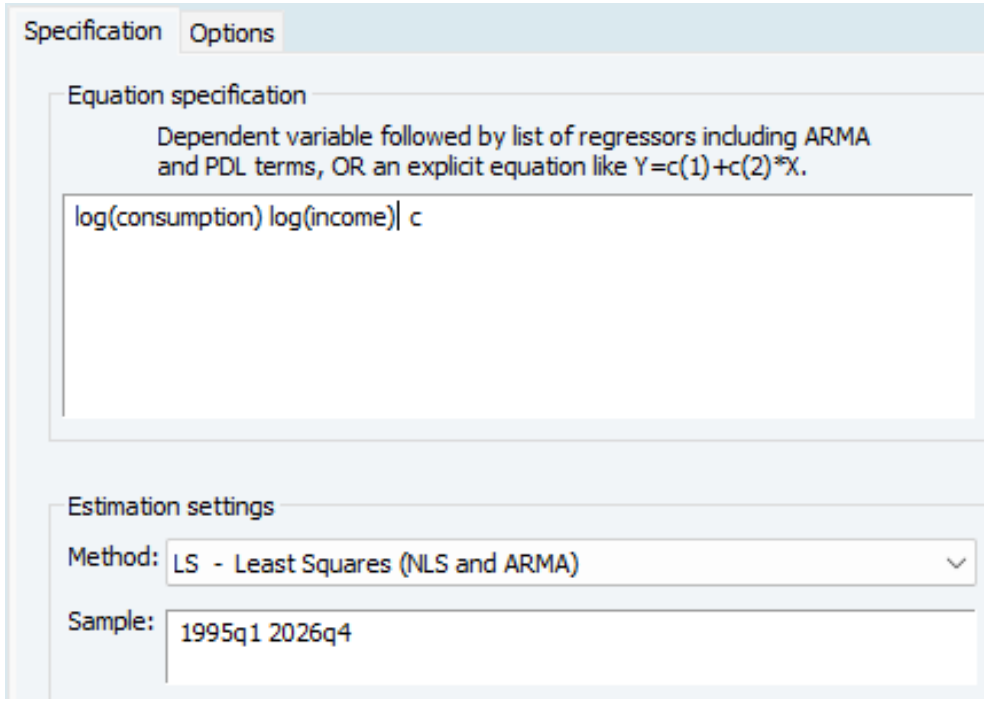
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(L_INC,2)
Method: Least Squares
Date: 08/12/26 Time: 16:54
Sample (adjusted): 1996Q1 2026Q2
Included observations: 122 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(L_INC(-1))	-1.119022	0.201860	-5.543570	0.0000
D(L_INC(-1),2)	-0.239046	0.151840	-1.574328	0.1181
D(L_INC(-2),2)	-0.319506	0.087662	-3.644750	0.0004
C	0.007144	0.001960	3.645678	0.0004

R-squared	0.722029	Mean dependent var	-7.74E-05
Adjusted R-squared	0.714962	S.D. dependent var	0.030010
S.E. of regression	0.016022	Akaike info criterion	-5.397492
Sum squared resid	0.030290	Schwarz criterion	-5.305557
Log likelihood	333.2470	Hannan-Quinn criter.	-5.360151
F-statistic	102.1683	Durbin-Watson stat	1.958492
Prob(F-statistic)	0.000000		

Step 2. Estimating the long-run relationship using OLS

The long-run multiplier has an expected sign and magnitude (close to 1), but we should test for cointegration



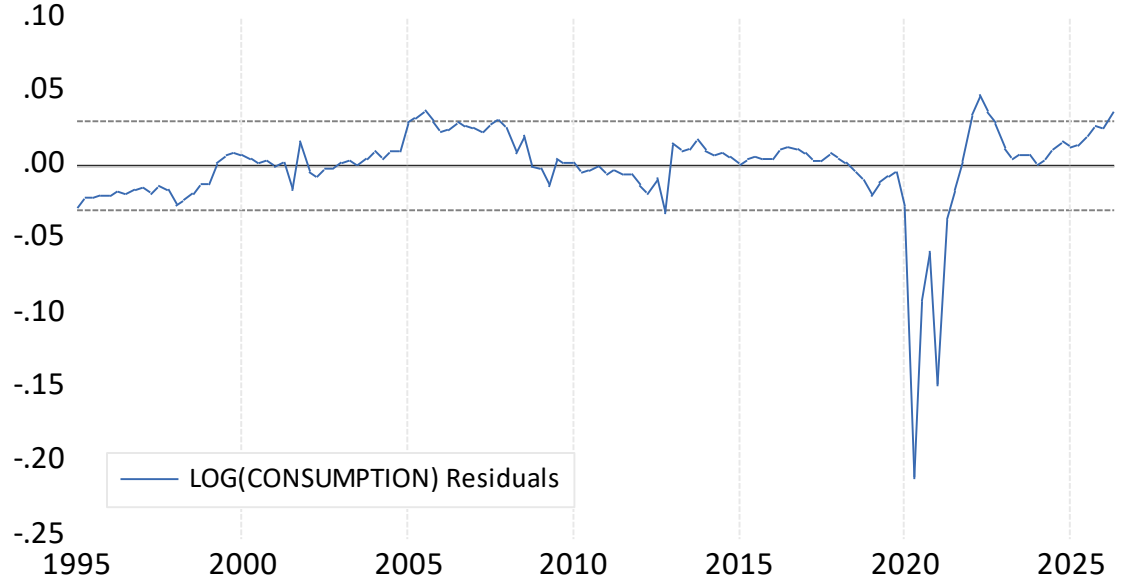
Dependent Variable: LOG(CONSUMPTION)
Method: Least Squares
Date: 08/12/26 Time: 16:59
Sample (adjusted): 1995Q1 2026Q2
Included observations: 126 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(INCOME)	0.975813	0.011860	82.28018	0.0000
C	0.130588	0.112007	1.165897	0.2459

R-squared	0.982013	Mean dependent var	9.343862
Adjusted R-squared	0.981868	S.D. dependent var	0.224161
S.E. of regression	0.030184	Akaike info criterion	-4.147258
Sum squared resid	0.112974	Schwarz criterion	-4.102238
Log likelihood	263.2773	Hannan-Quinn criter.	-4.128968
F-statistic	6770.028	Durbin-Watson stat	0.716931
Prob(F-statistic)	0.000000		

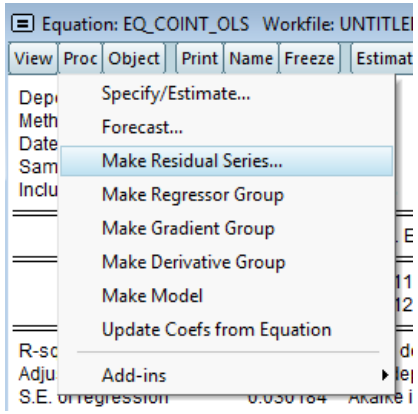
Step 3. Obtain residuals and test them for a unit-root (Engle-Granger test for cointegration)

Residuals of the long-run relationship show (zero) mean-reverting dynamics (the pandemic caused a huge temporary disequilibrium)



We reject the null of no cointegration at 5%:

t-stat = –3.89 < critical value = -3.40 (MacKinnon estimates)



Augmented Dickey-Fuller Unit Root Test on RESID_OLS

Null Hypothesis: RESID_OLS has a unit root
Exogenous: None
Lag Length: 3 (Automatic - based on SIC, maxlag=12)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.889172	0.0001
Test critical values:	1% level	-2.584055	
	5% level	-1.943471	
	10% level	-1.614984	

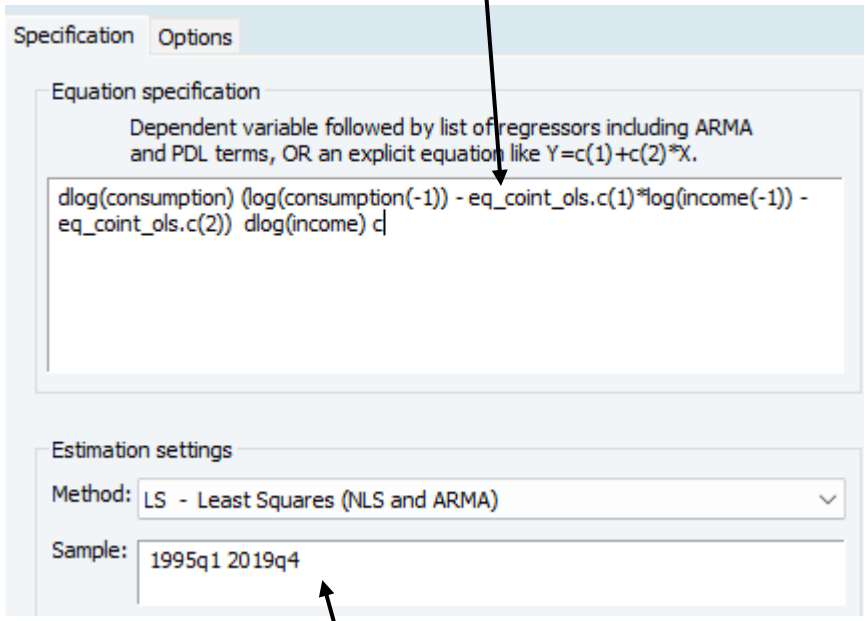
*MacKinnon (1996) one-sided p-values.

MacKinnon (2010) test critical values (100 observations, constant and no trend in the long-run equation)

No. of variables in the long-run	1%	5%	10%
2	-4.01	-3.40	-3.09
3	-4.44	-3.83	-3.51
4	-4.83	-4.21	-3.89

Step 4. Estimating an ECM

The lagged error term (from the long-run relationship)



We adjust the sample in order not to deal with the COVID-19 distorting influence.

The error-correction coefficient is significant and has the correct sign – if consumption grows above the equilibrium level it decreases next period

Dependent Variable: DLOG(CONSUMPTION)
Method: Least Squares
Date: 08/12/26 Time: 17:47
Sample (adjusted): 1995Q2 2019Q4
Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CONSUMPTION(-1))-EQ_COINT_OL...	-0.100230	0.030051	-3.335298	0.0012
DLOG(INCOME)	0.134390	0.051385	2.615343	0.0104
C	0.006089	0.000559	10.89543	0.0000

R-squared	0.136360	Mean dependent var	0.006763
Adjusted R-squared	0.118368	S.D. dependent var	0.004720
S.E. of regression	0.004432	Akaike info criterion	-7.970287
Sum squared resid	0.001885	Schwarz criterion	-7.891647
Log likelihood	397.5292	Hannan-Quinn criter.	-7.938469
F-statistic	7.578714	Durbin-Watson stat	1.135101
Prob(F-statistic)	0.000879		

The explanatory power of our ECM is low. We can add lags of consumption and income to account for the persistence.

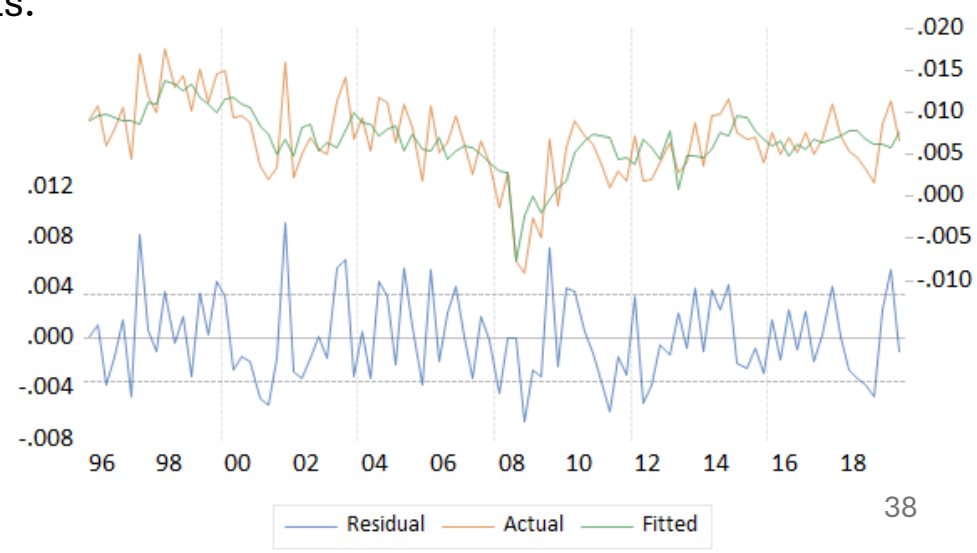
Step 4. Estimating an ECM: adjusting the model

Dependent Variable: DLOG(CONSUMPTION)
Method: Least Squares
Date: 08/12/26 Time: 17:56
Sample (adjusted): 1996Q1 2019Q4
Included observations: 96 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CONSUMPTION(-1))-EQ_COINT_OL...	-0.060951	0.025906	-2.352763	0.0208
DLOG(CONSUMPTION(-1))	0.249495	0.092300	2.703099	0.0082
DLOG(CONSUMPTION(-2))	0.170680	0.095689	1.783690	0.0779
DLOG(CONSUMPTION(-3))	0.236313	0.093062	2.539321	0.0128
DLOG(INCOME(-1))	0.075561	0.044795	1.686806	0.0951
C	0.002117	0.000730	2.901075	0.0047
D_2008Q3	-0.012205	0.003641	-3.351951	0.0012

R-squared	0.508551	Mean dependent var	0.006718
Adjusted R-squared	0.475420	S.D. dependent var	0.004784
S.E. of regression	0.003465	Akaike info criterion	-8.422309
Sum squared resid	0.001068	Schwarz criterion	-8.235325
Log likelihood	411.2708	Hannan-Quinn criter.	-8.346727
F-statistic	15.34953	Durbin-Watson stat	2.051845
Prob(F-statistic)	0.000000		

- We add lags to account for persistence (habit formation and consumption smoothing).
- The impulse dummy absorbs the effects of the Lehman Brothers collapse.
- The error-correction works. The error-correction coefficient is significant at 5% and has a correct sign: about 6% of the gap closes the next period.
- Residuals look like a white-noise, but we must check it using formal tests.



Step 4. Estimating an ECM: diagnostics. The ECM is adequate

Fail to reject the null of no residuals autocorrelation

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 4 lags

F-statistic	1.021499	Prob. F(4,85)	0.4010
Obs*R-squared	4.403112	Prob. Chi-Square(4)	0.3542

Test Equation:
Dependent Variable: RESID
Method: Least Squares
Date: 08/12/26 Time: 18:05
Sample: 1996Q1 2019Q4
Included observations: 96
Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CONSUMPTION(-1))-EQ_COINT_OL...	0.033361	0.034682	0.961902	0.3388
DLOG(CONSUMPTION(-1))	0.320888	0.245226	1.308540	0.1942
DLOG(CONSUMPTION(-2))	-0.086306	0.223973	-0.385341	0.7009
DLOG(CONSUMPTION(-3))	-0.124605	0.215740	-0.577569	0.5651
DLOG(INCOME(-1))	0.042614	0.050993	0.835682	0.4057
C	-0.001129	0.001108	-1.019320	0.3109
D_2008Q3	-0.000124	0.003651	-0.034014	0.9729
RESID(-1)	-0.381785	0.283924	-1.344675	0.1823
RESID(-2)	0.001152	0.240882	0.004783	0.9962
RESID(-3)	0.022721	0.195419	0.116266	0.9077
RESID(-4)	-0.248478	0.131295	-1.892523	0.0618

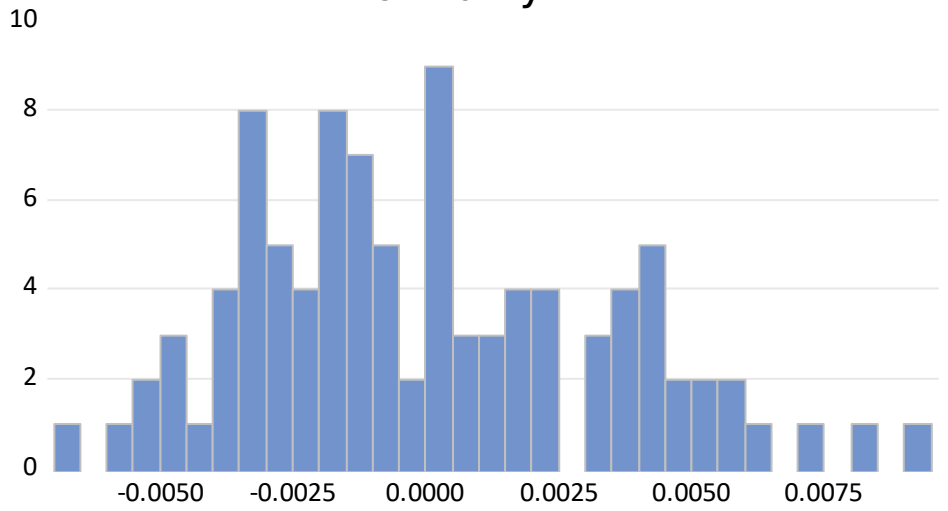
R-squared	0.045866	Mean dependent var	5.65E-20
Adjusted R-squared	-0.066385	S.D. dependent var	0.003353
S.E. of regression	0.003463	Akaike info criterion	-8.385926
Sum squared resid	0.001019	Schwarz criterion	-8.092095
Log likelihood	413.5245	Hannan-Quinn criter.	-8.267155
F-statistic	0.408600	Durbin-Watson stat	2.041712
Prob(F-statistic)	0.939069		

Fail to reject the null of residuals homoskedasticity

Heteroskedasticity Test: White
Null hypothesis: Homoskedasticity

F-statistic	1.107636	Prob. F(21,74)	0.3600
Obs*R-squared	22.95894	Prob. Chi-Square(21)	0.3462
Scaled explained SS	16.30850	Prob. Chi-Square(21)	0.7521

Fail to reject the null of residuals normality



Series: Residuals
Sample 1996Q1 2019Q4
Observations 96

Mean	5.65e-20
Median	-0.000631
Maximum	0.009167
Minimum	-0.006645
Std. Dev.	0.003353
Skewness	0.478193
Kurtosis	2.652931
Jarque-Bera	4.140526
Probability	0.126153

Step 4. Estimating an ECM: including the period after the COVID-19 pandemic began (1)

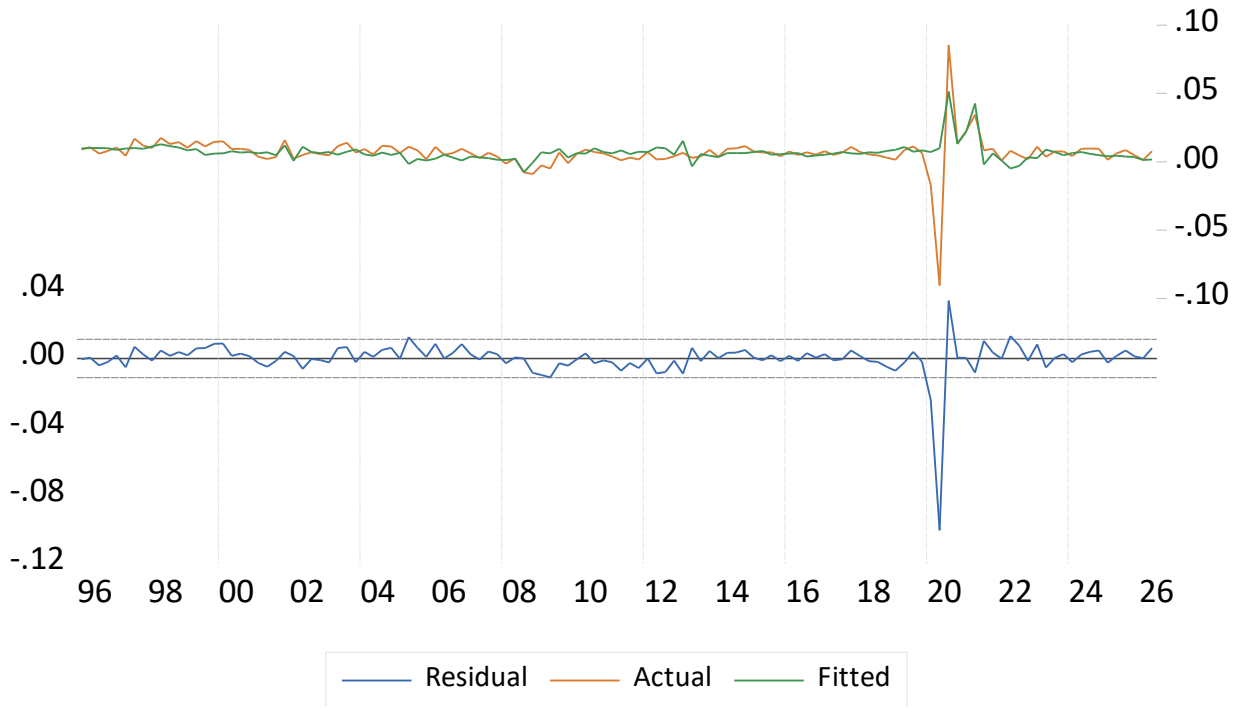
Now we use the whole sample from Q1-1995 to Q2-2026

Dependent Variable: DLOG(CONSUMPTION)
Method: Least Squares
Date: 08/12/26 Time: 19:49
Sample (adjusted): 1996Q1 2026Q2
Included observations: 122 after adjustments
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
RESID OLS(-1)	-0.152441	0.037696	-4.043993	0.0001
DLOG(CONSUMPTION(-1))	0.020708	0.121355	0.170642	0.8648
DLOG(CONSUMPTION(-2))	0.040800	0.066320	0.615199	0.5396
DLOG(CONSUMPTION(-3))	-0.081494	0.106873	-0.762533	0.4473
DLOG(INCOME(-1))	0.173963	0.123856	1.404553	0.1628
C	0.005893	0.001907	3.090597	0.0025
D2008 Q3	-0.015429	0.001701	-9.069724	0.0000

R-squared	0.264806	Mean dependent var	0.006716
Adjusted R-squared	0.226448	S.D. dependent var	0.012729
S.E. of regression	0.011196	Akaike info criterion	-6.090928
Sum squared resid	0.014414	Schwarz criterion	-5.930042
Log likelihood	378.5466	Hannan-Quinn criter.	-6.025581
F-statistic	6.903541	Durbin-Watson stat	2.021947
Prob(F-statistic)	0.000003	Wald F-statistic	151.3755
Prob(Wald F-statistic)	0.000000		

The problem: lockdowns and the pandemic-related support measures distort the error-correction mechanism



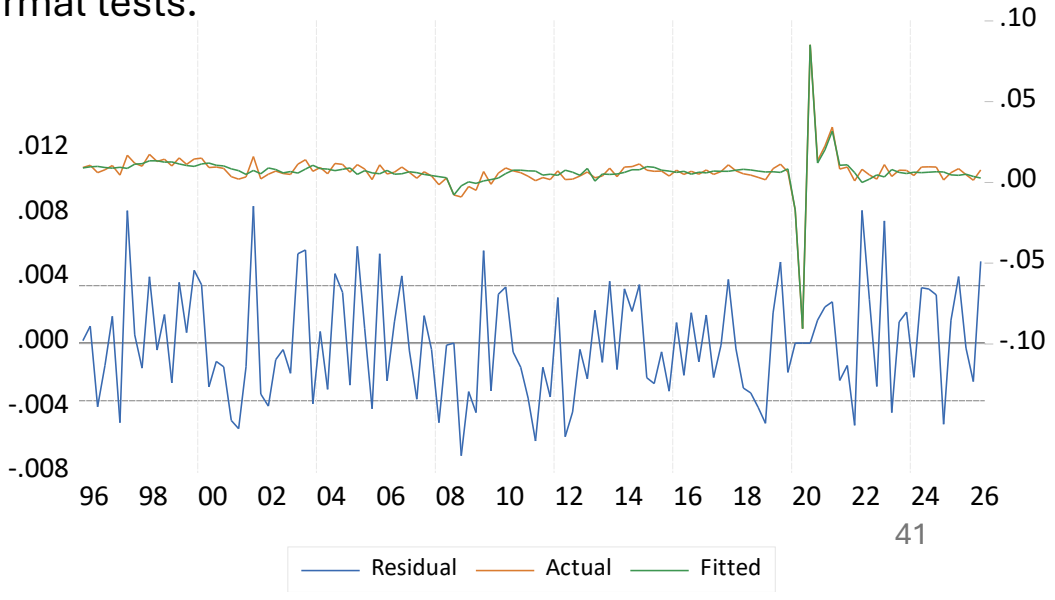
Step 4. Estimating an ECM: including the period after the COVID-19 pandemic began (2)

Dependent Variable: DLOG(CONSUMPTION)
Method: Least Squares
Date: 08/12/26 Time: 19:48
Sample (adjusted): 1996Q1 2026Q2
Included observations: 122 after adjustments
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
RESID_OLS(-1)	-0.057601	0.015189	-3.792318	0.0002
DLOG(CONSUMPTION(-1))	0.267115	0.052668	5.071702	0.0000
DLOG(CONSUMPTION(-2))	0.166719	0.022357	7.457016	0.0000
DLOG(CONSUMPTION(-3))	0.018695	0.023635	0.791023	0.4306
DLOG(INCOME(-1))	0.092398	0.025882	3.570021	0.0005
C	0.002721	0.000843	3.229009	0.0016
D2008_Q3	-0.012901	0.000599	-21.52997	0.0000
D2020_Q1	-0.024191	0.000398	-60.71799	0.0000
D2020_Q2	-0.092066	0.001739	-52.93910	0.0000
D2020_Q3	0.088524	0.007732	11.44933	0.0000
DLOG(CONSUMPTION(-3))*D_COVID	0.131930	0.059842	2.204624	0.0295

R-squared	0.928108	Mean dependent var	0.006716
Adjusted R-squared	0.921631	S.D. dependent var	0.012729
S.E. of regression	0.003563	Akaike info criterion	-8.350318
Sum squared resid	0.001410	Schwarz criterion	-8.097496
Log likelihood	520.3694	Hannan-Quinn criter.	-8.247630
F-statistic	143.2973	Durbin-Watson stat	2.190641
Prob(F-statistic)	0.000000		

- We add impulse dummies to account for 2020 lockdowns and transfers.
- We add the level-shift dummy to account for changing consumption smoothing pattern after 2019.
- The error-correction mechanism works, and the corresponding coefficient doesn't change much.
- Residuals look like a white-noise, but we must check it using formal tests.



Step 4. Estimating an ECM: including the period after the COVID-19 pandemic began (3)

Fail to reject the null of no residuals autocorrelation at 5% level

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 4 lags

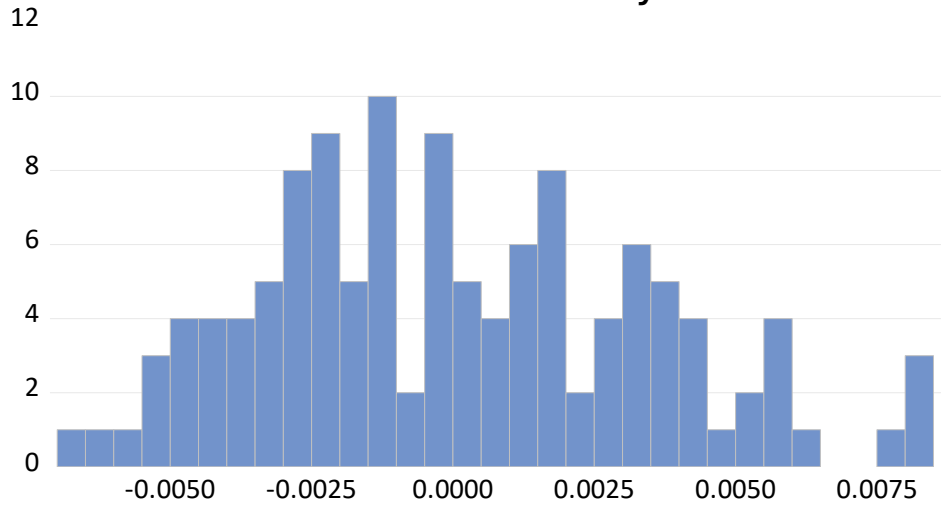
F-statistic	2.195425	Prob. F(4,107)	0.0743
Obs*R-squared	9.253341	Prob. Chi-Square(4)	0.0551

Fail to reject the null of residuals homoskedasticity at 5% level

Heteroskedasticity Test: White
Null hypothesis: Homoskedasticity

F-statistic	1.560638	Prob. F(30,91)	0.0559
Obs*R-squared	41.44514	Prob. Chi-Square(30)	0.0798
Scaled explained SS	25.88251	Prob. Chi-Square(30)	0.6811

Fail to reject the null of residuals normality

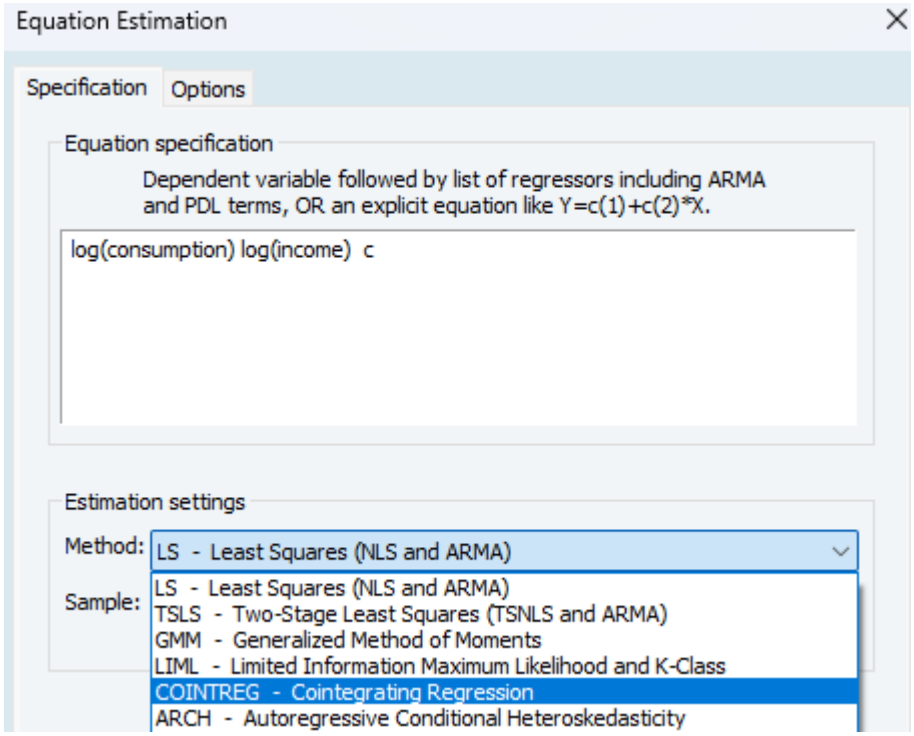


Series: Residuals
Sample 1996Q1 2026Q2
Observations 122

Mean	1.99e-19
Median	-0.000386
Maximum	0.008492
Minimum	-0.006961
Std. Dev.	0.003413
Skewness	0.355581
Kurtosis	2.508817
Jarque-Bera	3.797317
Probability	0.149769

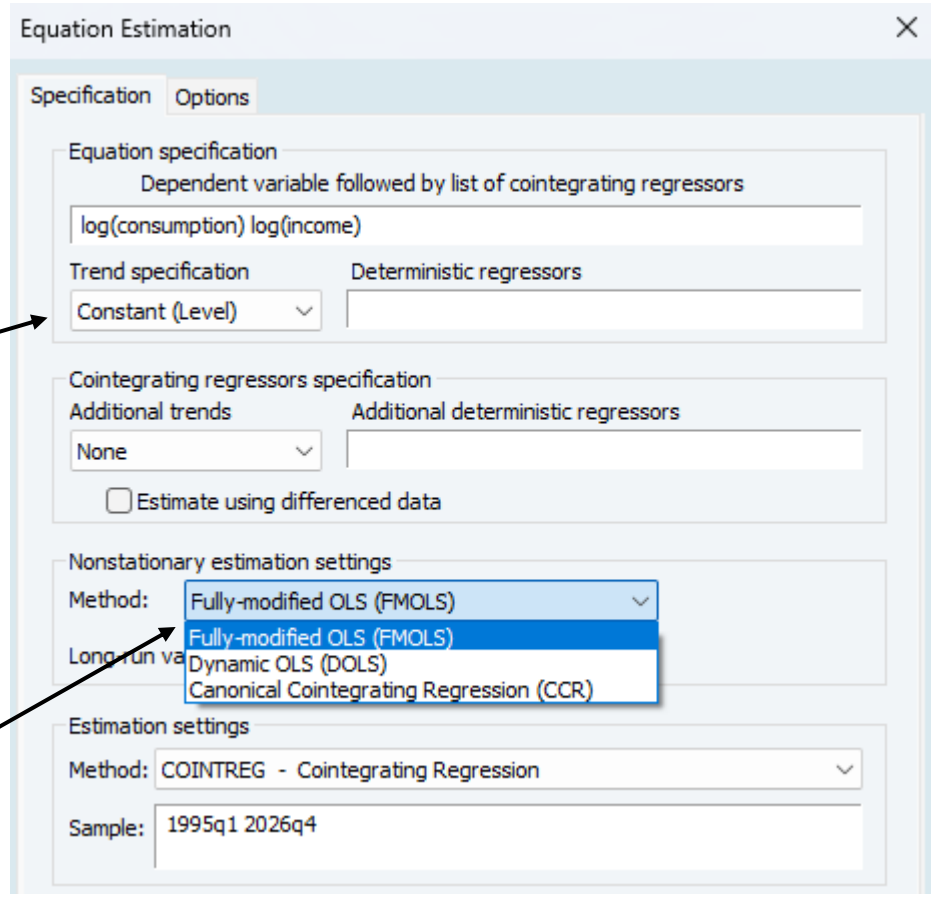
Alternative step 2. Using FMOLS/DOLS for a long-run relationship (1)

Select “Cointegrating Regression” method



Choose the terms to add to the cointegrating relation (a constant is the most common choice)

Choose the method



Alternative step 2. Using FMOLS/DOLS for a long-run relationship (2)

The long-run relationship (OLS)

Dependent Variable: LOG(CONSUMPTION)
Method: Least Squares
Date: 08/12/26 Time: 16:59
Sample (adjusted): 1995Q1 2026Q2
Included observations: 126 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(INCOME)	0.975813	0.011860	82.28018	0.0000
C	0.130588	0.112007	1.165897	0.2459

R-squared	0.982013	Mean dependent var	9.343862
Adjusted R-squared	0.981868	S.D. dependent var	0.224161
S.E. of regression	0.030184	Akaike info criterion	-4.147258
Sum squared resid	0.112974	Schwarz criterion	-4.102238
Log likelihood	263.2773	Hannan-Quinn criter.	-4.128968
F-statistic	6770.028	Durbin-Watson stat	0.716931
Prob(F-statistic)	0.000000		

The long-run relationship (FMOLS)

Dependent Variable: LOG(CONSUMPTION)
Method: Fully Modified Least Squares (FMOLS)
Date: 08/12/26 Time: 18:19
Sample (adjusted): 1995Q2 2026Q2
Included observations: 125 after adjustments
Cointegrating equation deterministics: C
Long-run covariance estimate (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(INCOME)	0.975621	0.020643	47.26224	0.0000
C	0.136553	0.195029	0.700167	0.4851

R-squared	0.981189	Mean dependent var	9.347524
Adjusted R-squared	0.981036	S.D. dependent var	0.221247
S.E. of regression	0.030468	Sum squared resid	0.114182
Long-run variance	0.002677		

The long-run relationship (DOLS)

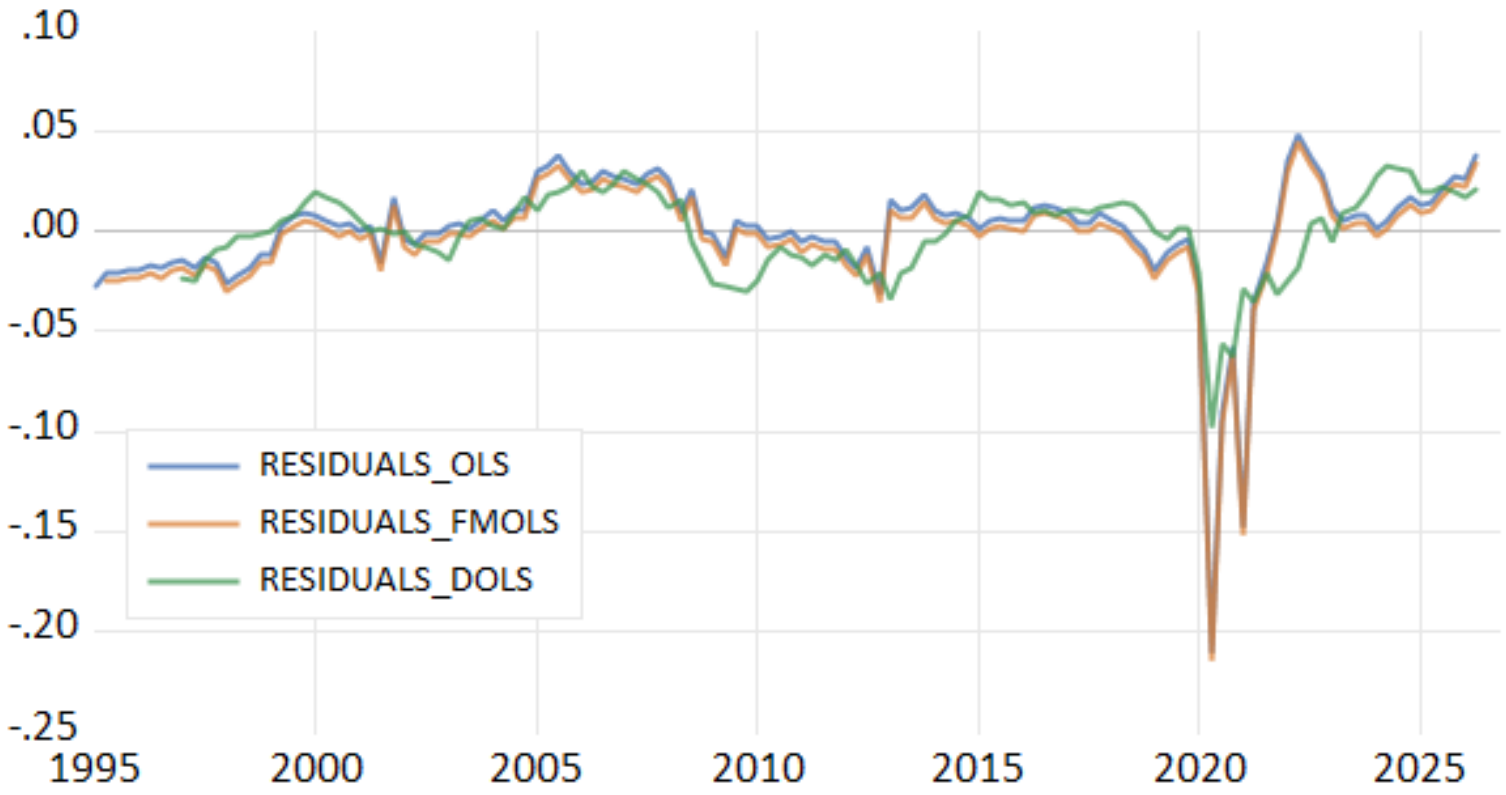
Dependent Variable: LOG(CONSUMPTION)
Method: Dynamic Least Squares (DOLS)
Date: 08/12/26 Time: 18:24
Sample (adjusted): 1997Q1 2026Q2
Included observations: 118 after adjustments
Cointegrating equation deterministics: C
Automatic leads and lags specification (lead=0 and lag=7 based on SIC criterion, max=12)
Long-run variance estimate (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(INCOME)	0.944810	0.019479	48.50447	0.0000
C	0.455670	0.185964	2.450310	0.0159

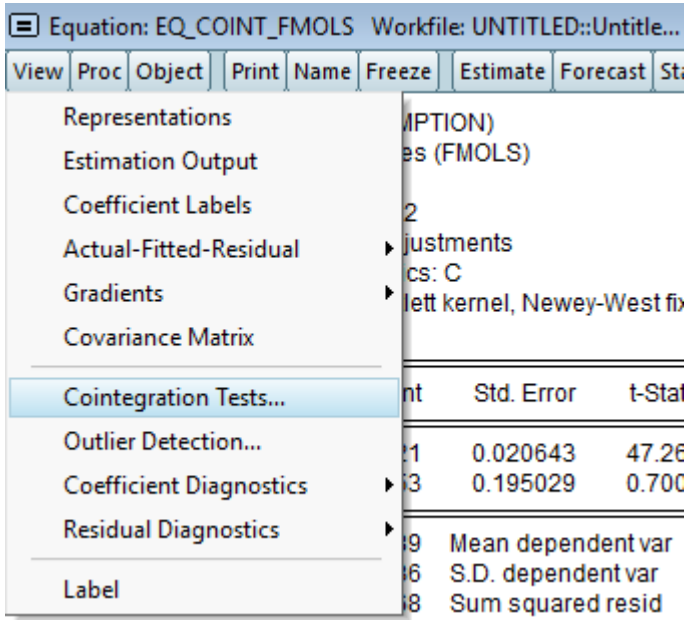
R-squared	0.989370	Mean dependent var	9.372870
Adjusted R-squared	0.988484	S.D. dependent var	0.200727
S.E. of regression	0.021541	Sum squared resid	0.050113
Long-run variance	0.001811		

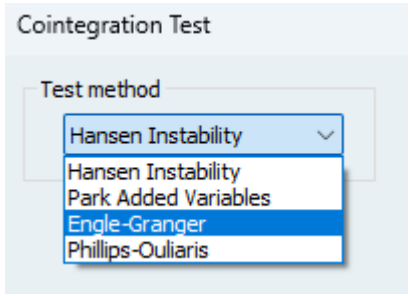
Compare slopes, not intercepts – β is the long-run content; the constant shifts mechanically with β.

Alternative step 2. Using FMOLS for a long-run relationship (3)



Alternative step 3. Testing cointegration directly from the “Equation” object





Cointegration Test - Engle-Granger
Date: 08/12/26 Time: 18:42
Equation: EQ_COINT_FMOLS
Specification: LOG(CONSUMPTION) LOG(INCOME) C
Cointegrating equation deterministics: C
Null hypothesis: Series are not cointegrated
Automatic lag specification (lag=3 based on Schwarz info criterion, maxlag=12)

	Value	Prob.*
Engle-Granger tau-statistic	-3.889172	0.0133
Engle-Granger z-statistic	-44.03540	0.0001

*MacKinnon (1996) p-values.

Intermediate Results:

Rho - 1	-0.308067
Rho S.E.	0.079211
Residual variance	0.000463
Long-run residual variance	0.000635
Number of lags	3
Number of observations	122
Number of stochastic trends**	2

**Number of stochastic trends in asymptotic distribution.

Engle-Granger Test Equation:
Dependent Variable: D(RESID)
Method: Least Squares
Date: 08/12/26 Time: 18:42
Sample (adjusted): 1996Q1 2026Q2
Included observations: 122 after adjustments

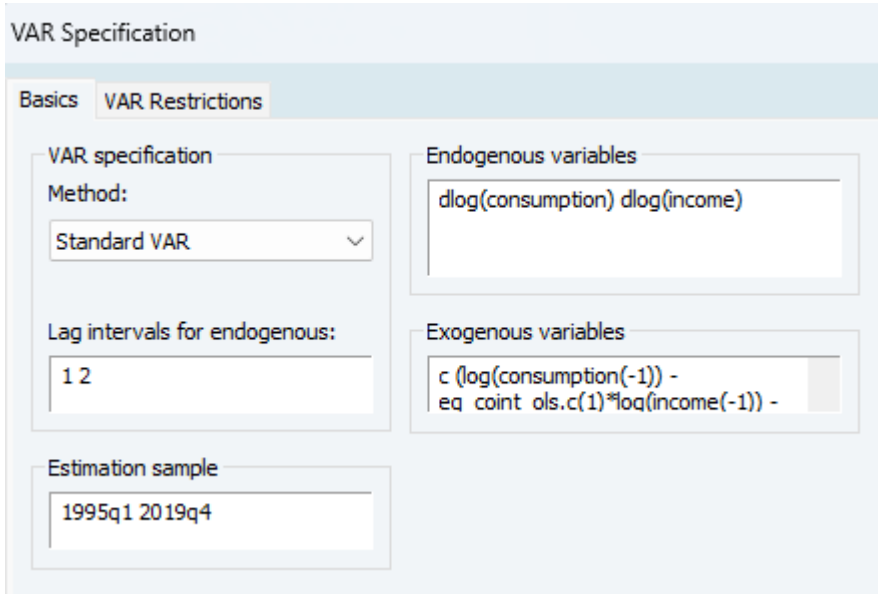
Variable	Coefficient	Std. Error	t-Statistic	Prob.
RESID(-1)	-0.308067	0.079211	-3.889172	0.0002
D(RESID(-1))	-0.132081	0.101486	-1.301471	0.1956
D(RESID(-2))	-0.051503	0.096384	-0.534358	0.5941
D(RESID(-3))	0.330086	0.087238	3.783742	0.0002

R-squared	0.325176	Mean dependent var	0.000471
Adjusted R-squared	0.308020	S.D. dependent var	0.025862
S.E. of regression	0.021513	Akaike info criterion	-4.808058
Sum squared resid	0.054613	Schwarz criterion	-4.716123
Log likelihood	297.2916	Hannan-Quinn criter.	-4.770717
Durbin-Watson stat	2.024565		

The Engle-Granger and Phillips-Ouliaris residual-based tests for cointegration are simply unit root tests applied to the residuals obtained from the static OLS estimation (not the FMOLS/DOLS).

Alternative step 4. Estimating the Vector Error-Correction Model (VECM)

Specify the VAR in first differences and add the lagged residuals (from the long-run regression) as an exogenous variable



Vector Autoregression Estimates
Date: 08/12/26 Time: 18:51
Sample (adjusted): 1995Q4 2019Q4
Included observations: 97 after adjustments
Standard errors in ( ) & t-statistics in [ ]

	DLOG(CON...	DLOG(INCOME)
DLOG(CONSUMPTION(...	0.343223	0.377036
	(0.09898)	(0.21434)
	[3.46762]	[1.75905]
DLOG(CONSUMPTION(...	0.314128	0.375413
	(0.09685)	(0.20972)
	[3.24355]	[1.79004]
DLOG(INCOME(-1))	0.040417	-0.353454
	(0.05214)	(0.11290)
	[0.77521]	[-3.13065]
DLOG(INCOME(-2))	-0.030641	0.014370
	(0.04990)	(0.10806)
	[-0.61401]	[0.13298]
C	0.002412	0.003771
	(0.00079)	(0.00172)
	[3.04022]	[2.19497]
LOG(CONSUMPTION(-1...	-0.066637	0.059341
	(0.02796)	(0.06054)
	[-2.38373]	[0.98025]
R-squared	0.410721	0.220432
Adj. R-squared	0.378343	0.177598
Sum sq. resids	0.001281	0.006007
S.E. equation	0.003752	0.008125
F-statistic	12.68519	5.146252
Log likelihood	407.2515	332.3040
Akaike AIC	-8.273226	-6.727917
Schwarz SC	-8.113966	-6.568656
Mean dependent	0.006720	0.006711
S.D. dependent	0.004759	0.008959

Correct sign and significant at 5%.

Correct sign but not significant – income is **weakly exogenous** (does not react to deviations from the long-run).

Alternative step 4. Estimating the VECM. Adjusting the model

Based on information criteria and model diagnostics we add lags of endogenous variables and the Lehman Brothers collapse dummy.

Residuals are not correlated and homoskedastic. The VAR is stationary.

VAR Residual Serial Correlation LM Tests
Date: 08/12/26 Time: 19:03
Sample: 1995Q1 2019Q4
Included observations: 95

Null hypothesis: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	2.737319	4	0.6027	0.685874	(4, 162.0)	0.6027
2	0.578243	4	0.9655	0.143929	(4, 162.0)	0.9655
3	2.643534	4	0.6191	0.662184	(4, 162.0)	0.6191
4	4.336158	4	0.3624	1.091847	(4, 162.0)	0.3624
5	0.798882	4	0.9386	0.198982	(4, 162.0)	0.9386
6	1.178014	4	0.8817	0.293757	(4, 162.0)	0.8817
7	6.191779	4	0.1853	1.568042	(4, 162.0)	0.1853
8	2.485446	4	0.6472	0.622282	(4, 162.0)	0.6473

Null hypothesis: No serial correlation at lags 1 to h

Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	2.737319	4	0.6027	0.685874	(4, 162.0)	0.6027
2	3.292817	8	0.9147	0.408092	(8, 158.0)	0.9147
3	5.992033	12	0.9165	0.492862	(12, 154.0)	0.9166
4	12.75871	16	0.6903	0.793678	(16, 150.0)	0.6910
5	15.17350	20	0.7664	0.750771	(20, 146.0)	0.7674
6	16.40237	24	0.8730	0.669545	(24, 142.0)	0.8741
7	23.49070	28	0.7081	0.829581	(28, 138.0)	0.7112
8	25.00319	32	0.8059	0.765089	(32, 134.0)	0.8093

*Edgeworth expansion corrected likelihood ratio statistic.

VAR Residual Heteroskedasticity Tests (Levels and Squares)
Date: 08/12/26 Time: 19:03
Sample: 1995Q1 2019Q4
Included observations: 95

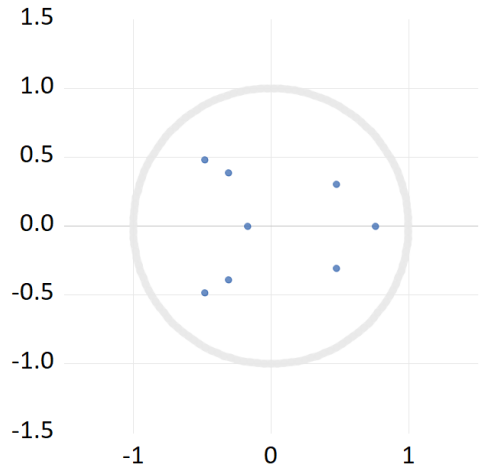
Joint test:

Chi-sq	df	Prob.
63.80921	57	0.2496

Individual components:

Dependent	R-squared	F(19,75)	Prob.	Chi-sq(19)	Prob.
res1*res1	0.143743	0.662658	0.8430	13.65556	0.8034
res2*res2	0.338542	2.020306	0.0169	32.16145	0.0300
res2*res1	0.173141	0.826560	0.6696	16.44835	0.6272

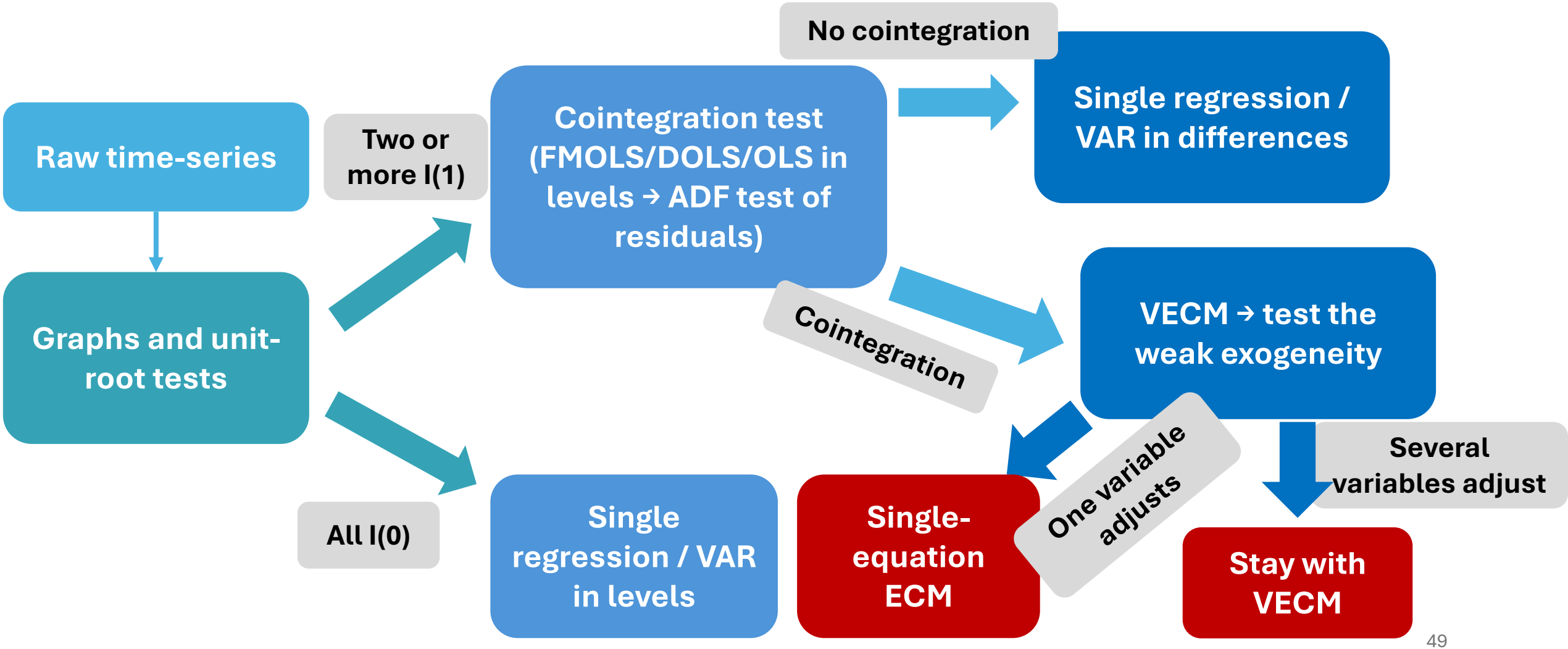
Inverse Roots of AR Characteristic Polynomial



Vector Autoregression Estimates
Date: 08/12/26 Time: 18:59
Sample (adjusted): 1996Q2 2019Q4
Included observations: 95 after adjustments
Standard errors in ( ) & t-statistics in [ ]

	DLOG(CON...	DLOG(INCOME)
DLOG(CONSUMPTION(...	0.271158	0.272468
	(0.09669)	(0.22090)
	[2.80426]	[1.23343]
DLOG(CONSUMPTION(...	0.240448	0.187217
	(0.10258)	(0.23436)
	[2.34392]	[0.79885]
DLOG(CONSUMPTION(...	0.345833	0.590113
	(0.10293)	(0.23514)
	[3.35994]	[2.50958]
DLOG(CONSUMPTION(...	-0.074547	-0.217734
	(0.10091)	(0.23053)
	[-0.73875]	[-0.94448]
DLOG(INCOME(-1))	0.052031	-0.307072
	(0.05222)	(0.11929)
	[0.99644]	[-2.57411]
DLOG(INCOME(-2))	-0.060376	0.032201
	(0.05258)	(0.12012)
	[-1.14832]	[0.26808]
DLOG(INCOME(-3))	-0.090801	-0.040978
	(0.05209)	(0.11900)
	[-1.74317]	[-0.34435]
DLOG(INCOME(-4))	-0.079274	-0.170010
	(0.04700)	(0.10738)
	[-1.68665]	[-1.58331]
C	0.003062	0.004441
	(0.00083)	(0.00189)
	[3.70984]	[2.35558]
LOG(CONSUMPTION(-1...	-0.082353	0.056671
	(0.02874)	(0.06565)
	[-2.86580]	[0.86323]
D_2008Q3	-0.011870	-0.019340
	(0.00366)	(0.00835)
	[-3.24606]	[-2.31508]
R-squared	0.541360	0.325973
Adj. R-squared	0.486760	0.245732
Sum sq. resids	0.000994	0.005189
S.E. equation	0.003440	0.007859
F-statistic	9.915031	4.062414

Summary.
Error-correction models estimation



Advantages of the ECM

- It measures the correction from last period's disequilibrium directly, and that coefficient has a clean economic reading: the share of the gap closed per period.
- It is estimated in first differences, so the trends are gone and the spurious-regression problem with them — but the level information survives in the error-correction term rather than being thrown away.
- The error term is stationary by construction, which is what guarantees the long-run errors cannot grow without bound.